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Page 1 of Lorem Ipsum Dolor Sit Amet Consectetur Adipiscing Elit Sed Do Eiusmod Tempor Incididunt Ut Labore Et Dolore Magna Aliqua Ut Enim Ad Minim Veniam Quis Nostrud Exercitation Ullamco Laboris Nisi Ut Aliquip Ex Ea Commodo Consequat Duis Aute Irure Dolor In Reprehenderit In Voluptate Velit Esse Cillum Dolore Eu Fugiat Nulla Pariatur Excepteur Sint Occaecat Cupidatat Non Proident Sunt In Culpa Qui Officia Deserunt Mollit Anim Id Est Laborum Sed Ut Perspiciatis Unde Omnis Iste Natus Error Sit Voluptatem Accusantium Doloremque Laudantium Totam Rem Aperiam Eaque Ipsa Quae Ab Illo Inventore Veritatis Et Quasi Architecto Beatae Vitae Dicta Sunt Explicabo Nemo Enim Ipsam Voluptatem Quia Voluptas Sit Aspernatur Aut Odit Aut Fugit Sed Quia Consequuntur Magni Dolores Eos Qui Ratione Voluptatem Sequi Nesciunt Neque Porro Quisquam Est Qui Dolorem Ipsum Quia Dolor Sit Amet Consectetur Adipisci Velit Sed Quia Non Numquam Eius Modi Tempora Incidunt Ut Labore Et Dolore Magnam Aliquam Quaerat Voluptatem Ut Enim Ad Minima Veniam Quis Nostrum Exercitationem Ullam Corporis Suscipit Laboriosam Nisi Ut Aliquid Ex Ea Commodi Consequatur Quis Autem Vel Eum Iure Reprehenderit Qui In Ea Voluptate Velit Esse Quam Nihil Molestiae Consequatur Vel Illum Qui Dolorem Eum Fugiat Quo Voluptas Nulla Pariatur At Vero Eos Et Accusamus Et Iusto Odio Dignissimos Ducimus Qui Blanditiis Praesentium Voluptatum Deleniti Atque Corrupti Quos Dolores Et Quas Molestias Excepturi Sint Occaecati Cupiditate Non Provident Similique Sunt In Culpa Qui Officia Deserunt Mollitia Animi Id Est Laborum Et Dolorum Fuga Et Harum Quidem Rerum Facilis Est Et Expedita Distinctio Nam Libero Tempore Cum Soluta Nobis Est Eligendi Optio Cumque Nihil Impedit Quo Minus Id Quod Maxi Placeat Facere Possimus Omnis Voluptas Assumenda Est Omnis Dolor Repellendus Temporibus Autem Quibusdam Et Aut Officiis Debitis Aut Rerum Necessitatibus Saepe Eveniet Ut Et Voluptates Repudiandae Sint Et Molestiae Non Recusandae Itaque Earum Rerum Hic Tenetur A Sapiente Delectus Ut Aut Reiciendis Voluptatibus Maiores Alias Consequatur Aut Perferendis Doloribus Asperiores Repellat

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Effect of the cities on an introduction of a language and Mandarin Accents - The Mandarin Speech Accent of a LCS in central Appalachian of Pennsylvania, an ex-Coal Town, and Industrial Landscape

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Abstract

Mandarin Accents of speakers, the effects of immigration, acculturation and adaptation of immigrant speakers, and the influences of a speaker's native and other languages on the accents of speakers of Mandarin are discussed. The Mandarin Speech Accent of a LCS in central Appalachian of Pennsylvania, an ex-Coal Town, and Industrial Landscape, Mandarin Speech Accent of a Native Speaker of Chinese Language, Mandarin Speech Accent of a Native Speaker of Japanese Language, Mandarin Speech Accent of a Native Speaker of Korean Language, Mandarin Speech Accent of a Native Speaker of Vietnamese Language, Mandarin Speech Accent of a Native Speaker of Thai Language, Mandarin Speech Accent of a Native Speaker of Tagalog Language, Mandarin Speech Accent of a Native Speaker of Hindi Language, Mandarin Speech Accent of a Native Speaker of Bengali Language, Mandarin Speech Accent of a Native Speaker of Punjabi Language, Mandarin Speech Accent of a Native Speaker of Urdu Language, Mandarin Speech Accent of a Native Speaker of Gujarati Language, Mandarin Speech Accent of a Native Speaker of Telugu Language, Mandarin Speech Accent of a Native Speaker of Tamil Language, Mandarin Speech Accent of a Native Speaker of Malayalam Language, Mandarin Speech Accent of a Native Speaker of Kannada Language, Mandarin Speech Accent of a Native Speaker of Marathi Language, Mandarin Speech Accent of a Native Speaker of Sindhi Language, Mandarin Speech Accent of a Native Speaker of Pashto Language, Mandarin Speech Accent of a Native Speaker of Dari Language, Mandarin Speech Accent of a Native Speaker of Persian Language, Mandarin Speech Accent of a Native Speaker of Arabic Language, Mandarin Speech Accent of a Native Speaker of Hebrew Language, Mandarin Speech Accent of a Native Speaker of Yiddish Language, Mandarin Speech Accent of a Native Speaker of Russian Language, Mandarin Speech Accent of a Native Speaker of Ukrainian Language, Mandarin Speech Accent of a Native Speaker of Polish Language, Mandarin Speech Accent of a Native Speaker of Czech Language, Mandarin Speech Accent of a Native Speaker of Slovak Language, Mandarin Speech Accent of a Native Speaker of Hungarian Language, Mandarin Speech Accent of a Native Speaker of Romanian Language, Mandarin Speech Accent of a Native Speaker of Bulgarian Language, Mandarin Speech Accent of a Native Speaker of Serbian Language, Mandarin Speech Accent of a Native Speaker of Croatian Language, Mandarin Speech Accent of a Native Speaker of Bosnian Language, Mandarin Speech Accent of a Native Speaker of Slovenian Language, Mandarin Speech Accent of a Native Speaker of Greek Language, Mandarin Speech Accent of a Native Speaker of Albanian Language, Mandarin Speech Accent of a Native Speaker of Turkish Language, Mandarin Speech Accent of a Native Speaker of Swahili Language, Mandarin Speech Accent of a Native Speaker of Hausa Language, Mandarin Speech Accent of a Native Speaker of Yoruba Language, Mandarin Speech Accent of a Native Speaker of Igbo Language, Mandarin Speech Accent of a Native Speaker of Zulu Language, Mandarin Speech Accent of a Native Speaker of Xhosa Language, Mandarin Speech Accent of a Native Speaker of Afrikaans Language, Mandarin Speech Accent of a Native Speaker of Sesotho Language, Mandarin Speech Accent of a Native Speaker of Tswana Language, Mandarin Speech Accent of a Native Speaker of Sesotho Language, Mandarin Speech Accent of a Native Speaker of Tswana

Name	Product Type	Status
Cable Assembly, 2 Conductor, Jumper, Jumper, Jumper Cable, Jumper Cable, Jumper Cable

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{"id":"1231","title":"Ultra-wide Lumbar ","
"Lumbar Spine","
"Lumbar Spine","
"Lumbar Spine
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1.1 Different forms and forces

Forces

Forces are pushes or pulls. **Forces** sometimes **cause** an **object** to **move**, or **change** its **speed** or **direction**. **Forces** can also make an **object** **stretch**, **bend** or **twist**. **Forces** can also **stop** a **moving** **object**. **Forces** **cannot** **make** **something** **appear** or **disappear**.

Force

Force (noun) forces, forces, forces,

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Proof. Let $\mathbf{A}$ be a matrix satisfying the regularity condition (1.1.1). Then $\mathbf{A}$ is invertible and satisfies condition (1.1.2).

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Ans. $\sqrt{\cos\theta}\,\sin\theta\,\sqrt{\sin\theta}\,\sin\theta\,\sqrt{\cos\theta}\,\cos\theta$

Ans. $\sqrt{\cos\theta}\,\sin\theta\,\sqrt{\sin\theta}\,\sin\theta\,\sqrt{\cos\theta}\,\cos\theta$

Ans.

$$\int_0^{\pi/2}\sqrt{\cos\theta}\,\sin\theta\,\sqrt{\sin\theta}\,\sin\theta\,\sqrt{\cos\theta}\,\cos\theta\,d\theta=\int_0^{\pi/2}\sin^2\theta\cos^2\theta\,d\theta=\int_0^{\pi/2}\frac{1}{4}\sin^2 2\theta\,d\theta=\frac{1}{8}\int_0^{\pi/2}(1-\cos 4\theta)\,d\theta=\frac{\pi}{16}$$

Ans. $\frac{\pi}{16}$

Ans. $\frac{\pi}{16}$

Ans. $\frac{\pi}{16}

Exercise Textbook exercise 2.2 (Pg. 50)

Part a Show that the sum of two odd integers is even.

Proof Let m and n be odd integers. Then $m = 2k_1 + 1$ and $n = 2k_2 + 1$ for some integers k_1 and k_2. Then $m + n = 2(k_1 + k_2 + 1)$. Since $k_1 + k_2 + 1$ is an integer, $m + n$ is even.

Part b Show that the product of two odd integers is odd.

Proof Let m and n be odd integers. Then $m = 2k_1 + 1$ and $n = 2k_2 + 1$ for some integers k_1 and k_2. Then $mn = 2(2k_1k_2 + k_1 + k_2) + 1$. Since $2k_1k_2 + k_1 + k_2$ is an integer, mn is odd.

Exercise Textbook exercise 2.3 (Pg. 50)

Proposition If n is an integer, $n^2 + 1$ is odd if and only if n is even.

Proof Let n be an integer.

Part a If n is even, then $n^2 + 1$ is odd. Suppose n is even. Then $n = 2k$ for some integer k. Then $n^2 + 1 = 2(2k^2) + 1$. Since $2k^2$ is an integer, $n^2 + 1$ is odd.

Part b If $n^2 + 1$ is odd, then n is even. Suppose $n^2 + 1$ is odd. Then $n^2 = 2k$ for some integer k. Then $n = \sqrt{2k}$. Since $\sqrt{2k}$ is an integer, n is even.

Exercise Textbook exercise 2.4 (Pg. 50)

Proposition If n is an integer, n^2 is even if and only if n is even.

Proof Let n be an integer.

Part a If n is even, then n^2 is even. Suppose n is even. Then $n = 2k$ for some integer k. Then $n^2 = 2(2k^2)$. Since $2k^2$ is an integer, n^2 is even.

Part b If n^2 is even, then n is even. Suppose n^2 is even. Then $n^2 = 2k$ for some integer k. Then $n = \sqrt{2k}$. Since $\sqrt{2k}$ is an integer, n is even.

Exercise Textbook exercise 2.5 (Pg. 50)

Proposition If n is an integer, n^2 is odd if and only if n is odd.

Proof Let n be an integer.

Part a If n is odd, then n^2 is odd. Suppose n is odd. Then $n = 2k + 1$ for some integer k. Then $n^2 = 2(2k^2 + 2k) + 1$. Since $2k^2 + 2k$ is an integer, n^2 is odd.

Part b If n^2 is odd, then n is odd. Suppose n^2 is odd. Then $n^2 = 2k + 1$ for some integer k. Then $n = \sqrt{2k + 1}$. Since $\sqrt{2k + 1}$ is an integer, n is odd.

Exercise Textbook exercise 2.6 (Pg. 50)

Proposition If n is an integer, $n^2 + n$ is even.

Proof Let n be an integer. Then $n^2 + n = n(n + 1)$. Since n or $n + 1$ is even, $n^2 + n$ is even.

Exercise Textbook exercise 2.7 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 2$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 2 = (n + 1)(n + 2)$. Since $n + 1$ or $n + 2$ is even, $n^2 + 3n + 2$ is even.

Exercise Textbook exercise 2.8 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 4$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 4 = (n + 1)(n + 2) + 2$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 4$ is odd.

Exercise Textbook exercise 2.9 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 5$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 5 = (n + 1)(n + 2) + 3$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 5$ is odd.

Exercise Textbook exercise 2.10 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 6$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 6 = (n + 1)(n + 2) + 4$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 6$ is even.

Exercise Textbook exercise 2.11 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 7$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 7 = (n + 1)(n + 2) + 5$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 7$ is odd.

Exercise Textbook exercise 2.12 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 8$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 8 = (n + 1)(n + 2) + 6$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 8$ is even.

Exercise Textbook exercise 2.13 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 9$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 9 = (n + 1)(n + 2) + 7$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 9$ is odd.

Exercise Textbook exercise 2.14 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 10$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 10 = (n + 1)(n + 2) + 8$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 10$ is even.

Exercise Textbook exercise 2.15 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 11$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 11 = (n + 1)(n + 2) + 9$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 11$ is odd.

Exercise Textbook exercise 2.16 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 12$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 12 = (n + 1)(n + 2) + 10$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 12$ is even.

Exercise Textbook exercise 2.17 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 13$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 13 = (n + 1)(n + 2) + 11$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 13$ is odd.

Exercise Textbook exercise 2.18 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 14$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 14 = (n + 1)(n + 2) + 12$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 14$ is even.

Exercise Textbook exercise 2.19 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 15$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 15 = (n + 1)(n + 2) + 13$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 15$ is odd.

Exercise Textbook exercise 2.20 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 16$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 16 = (n + 1)(n + 2) + 14$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 16$ is even.

Exercise Textbook exercise 2.21 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 17$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 17 = (n + 1)(n + 2) + 15$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 17$ is odd.

Exercise Textbook exercise 2.22 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 18$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 18 = (n + 1)(n + 2) + 16$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 18$ is even.

Exercise Textbook exercise 2.23 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 19$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 19 = (n + 1)(n + 2) + 17$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 19$ is odd.

Exercise Textbook exercise 2.24 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 20$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 20 = (n + 1)(n + 2) + 18$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 20$ is even.

Exercise Textbook exercise 2.25 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 21$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 21 = (n + 1)(n + 2) + 19$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 21$ is odd.

Exercise Textbook exercise 2.26 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 22$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 22 = (n + 1)(n + 2) + 20$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 22$ is even.

Exercise Textbook exercise 2.27 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 23$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 23 = (n + 1)(n + 2) + 21$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 23$ is odd.

Exercise Textbook exercise 2.28 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 24$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 24 = (n + 1)(n + 2) + 22$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 24$ is even.

Exercise Textbook exercise 2.29 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 25$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 25 = (n + 1)(n + 2) + 23$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 25$ is odd.

Exercise Textbook exercise 2.30 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 26$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 26 = (n + 1)(n + 2) + 24$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 26$ is even.

Exercise Textbook exercise 2.31 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 27$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 27 = (n + 1)(n + 2) + 25$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 27$ is odd.

Exercise Textbook exercise 2.32 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 28$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 28 = (n + 1)(n + 2) + 26$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 28$ is even.

Exercise Textbook exercise 2.33 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 29$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 29 = (n + 1)(n + 2) + 27$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 29$ is odd.

Exercise Textbook exercise 2.34 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 30$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 30 = (n + 1)(n + 2) + 28$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 30$ is even.

Exercise Textbook exercise 2.35 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 31$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 31 = (n + 1)(n + 2) + 29$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 31$ is odd.

Exercise Textbook exercise 2.36 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 32$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 32 = (n + 1)(n + 2) + 30$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 32$ is even.

Exercise Textbook exercise 2.37 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 33$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 33 = (n + 1)(n + 2) + 31$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 33$ is odd.

Exercise Textbook exercise 2.38 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 34$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 34 = (n + 1)(n + 2) + 32$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 34$ is even.

Exercise Textbook exercise 2.39 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 35$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 35 = (n + 1)(n + 2) + 33$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 35$ is odd.

Exercise Textbook exercise 2.40 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 36$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 36 = (n + 1)(n + 2) + 34$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 36$ is even.

Exercise Textbook exercise 2.41 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 37$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 37 = (n + 1)(n + 2) + 35$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 37$ is odd.

Exercise Textbook exercise 2.42 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 38$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 38 = (n + 1)(n + 2) + 36$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 38$ is even.

Exercise Textbook exercise 2.43 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 39$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 39 = (n + 1)(n + 2) + 37$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 39$ is odd.

Exercise Textbook exercise 2.44 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 40$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 40 = (n + 1)(n + 2) + 38$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 40$ is even.

Exercise Textbook exercise 2.45 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 41$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 41 = (n + 1)(n + 2) + 39$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 41$ is odd.

Exercise Textbook exercise 2.46 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 42$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 42 = (n + 1)(n + 2) + 40$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 42$ is even.

Exercise Textbook exercise 2.47 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 43$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 43 = (n + 1)(n + 2) + 41$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 43$ is odd.

Exercise Textbook exercise 2.48 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 44$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 44 = (n + 1)(n + 2) + 42$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 44$ is even.

Exercise Textbook exercise 2.49 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 45$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 45 = (n + 1)(n + 2) + 43$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 45$ is odd.

Exercise Textbook exercise 2.50 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 46$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 46 = (n + 1)(n + 2) + 44$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 46$ is even.

Exercise Textbook exercise 2.51 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 47$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 47 = (n + 1)(n + 2) + 45$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 47$ is odd.

Exercise Textbook exercise 2.52 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 48$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 48 = (n + 1)(n + 2) + 46$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 48$ is even.

Exercise Textbook exercise 2.53 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 49$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 49 = (n + 1)(n + 2) + 47$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 49$ is odd.

Exercise Textbook exercise 2.54 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 50$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 50 = (n + 1)(n + 2) + 48$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 50$ is even.

Exercise Textbook exercise 2.55 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 51$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 51 = (n + 1)(n + 2) + 49$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 51$ is odd.

Exercise Textbook exercise 2.56 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 52$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 52 = (n + 1)(n + 2) + 50$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 52$ is even.

Exercise Textbook exercise 2.57 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 53$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 53 = (n + 1)(n + 2) + 51$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 53$ is odd.

Exercise Textbook exercise 2.58 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 54$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 54 = (n + 1)(n + 2) + 52$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 54$ is even.

Exercise Textbook exercise 2.59 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 55$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 55 = (n + 1)(n + 2) + 53$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 55$ is odd.

Exercise Textbook exercise 2.60 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 56$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 56 = (n + 1)(n + 2) + 54$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 56$ is even.

Exercise Textbook exercise 2.61 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 57$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 57 = (n + 1)(n + 2) + 55$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 57$ is odd.

Exercise Textbook exercise 2.62 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 58$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 58 = (n + 1)(n + 2) + 56$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 58$ is even.

Exercise Textbook exercise 2.63 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 59$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 59 = (n + 1)(n + 2) + 57$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 59$ is odd.

Exercise Textbook exercise 2.64 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 60$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 60 = (n + 1)(n + 2) + 58$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 60$ is even.

Exercise Textbook exercise 2.65 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 61$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 61 = (n + 1)(n + 2) + 59$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 61$ is odd.

Exercise Textbook exercise 2.66 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 62$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 62 = (n + 1)(n + 2) + 60$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 62$ is even.

Exercise Textbook exercise 2.67 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 63$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 63 = (n + 1)(n + 2) + 61$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 63$ is odd.

Exercise Textbook exercise 2.68 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 64$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 64 = (n + 1)(n + 2) + 62$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 64$ is even.

Exercise Textbook exercise 2.69 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 65$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 65 = (n + 1)(n + 2) + 63$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 65$ is odd.

Exercise Textbook exercise 2.70 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 66$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 66 = (n + 1)(n + 2) + 64$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 66$ is even.

Exercise Textbook exercise 2.71 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 67$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 67 = (n + 1)(n + 2) + 65$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 67$ is odd.

Exercise Textbook exercise 2.72 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 68$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 68 = (n + 1)(n + 2) + 66$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 68$ is even.

Exercise Textbook exercise 2.73 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 69$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 69 = (n + 1)(n + 2) + 67$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 69$ is odd.

Exercise Textbook exercise 2.74 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 70$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 70 = (n + 1)(n + 2) + 68$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 70$ is even.

Exercise Textbook exercise 2.75 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 71$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 71 = (n + 1)(n + 2) + 69$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 71$ is odd.

Exercise Textbook exercise 2.76 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 72$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 72 = (n + 1)(n + 2) + 70$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 72$ is even.

Exercise Textbook exercise 2.77 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 73$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 73 = (n + 1)(n + 2) + 71$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 73$ is odd.

Exercise Textbook exercise 2.78 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 74$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 74 = (n + 1)(n + 2) + 72$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 74$ is even.

Exercise Textbook exercise 2.79 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 75$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 75 = (n + 1)(n + 2) + 73$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 75$ is odd.

Exercise Textbook exercise 2.80 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 76$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 76 = (n + 1)(n + 2) + 74$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 76$ is even.

Exercise Textbook exercise 2.81 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 77$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 77 = (n + 1)(n + 2) + 75$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 77$ is odd.

Exercise Textbook exercise 2.82 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 78$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 78 = (n + 1)(n + 2) + 76$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 78$ is even.

Exercise Textbook exercise 2.83 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 79$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 79 = (n + 1)(n + 2) + 77$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 79$ is odd.

Exercise Textbook exercise 2.84 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 80$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 80 = (n + 1)(n + 2) + 78$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 80$ is even.

Exercise Textbook exercise 2.85 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 81$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 81 = (n + 1)(n + 2) + 79$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 81$ is odd.

Exercise Textbook exercise 2.86 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 82$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 82 = (n + 1)(n + 2) + 80$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 82$ is even.

Exercise Textbook exercise 2.87 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 83$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 83 = (n + 1)(n + 2) + 81$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 83$ is odd.

Exercise Textbook exercise 2.88 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 84$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 84 = (n + 1)(n + 2) + 82$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 84$ is even.

Exercise Textbook exercise 2.89 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 85$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 85 = (n + 1)(n + 2) + 83$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 85$ is odd.

Exercise Textbook exercise 2.90 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 86$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 86 = (n + 1)(n + 2) + 84$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 86$ is even.

Exercise Textbook exercise 2.91 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 87$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 87 = (n + 1)(n + 2) + 85$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 87$ is odd.

Exercise Textbook exercise 2.92 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 88$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 88 = (n + 1)(n + 2) + 86$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 88$ is even.

Exercise Textbook exercise 2.93 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 89$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 89 = (n + 1)(n + 2) + 87$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 89$ is odd.

Exercise Textbook exercise 2.94 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 90$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 90 = (n + 1)(n + 2) + 88$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 90$ is even.

Exercise Textbook exercise 2.95 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 91$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 91 = (n + 1)(n + 2) + 89$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 91$ is odd.

Exercise Textbook exercise 2.96 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 92$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 92 = (n + 1)(n + 2) + 90$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 92$ is even.

Exercise Textbook exercise 2.97 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 93$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 93 = (n + 1)(n + 2) + 91$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 93$ is odd.

Exercise Textbook exercise 2.98 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 94$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 94 = (n + 1)(n + 2) + 92$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 94$ is even.

Exercise Textbook exercise 2.99 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 95$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 95 = (n + 1)(n + 2) + 93$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 95$ is odd.

Exercise Textbook exercise 2.100 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 96$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 96 = (n + 1)(n + 2) + 94$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 96$ is even.

Exercise Textbook exercise 2.101 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 97$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 97 = (n + 1)(n + 2) + 95$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 97$ is odd.

Exercise Textbook exercise 2.102 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 98$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 98 = (n + 1)(n + 2) + 96$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 98$ is even.

Exercise Textbook exercise 2.103 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 99$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 99 = (n + 1)(n + 2) + 97$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 99$ is odd.

Exercise Textbook exercise 2.104 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 100$ is even.

Proof Let n be an integer. Then $n^2 + 3n + 100 = (n + 1)(n + 2) + 98$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 100$ is even.

Exercise Textbook exercise 2.105 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 101$ is odd.

Proof Let n be an integer. Then $n^2 + 3n + 101 = (n + 1)(n + 2) + 99$. Since $(n + 1)(n + 2)$ is even, $n^2 + 3n + 101$ is odd.

Exercise Textbook exercise 2.106 (Pg. 50)

Proposition If n is an integer, $n^2 + 3n + 102$ is even.

Proof Let n be an integer

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    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1237",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1237",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1238",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1238",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1239",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1239",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1240",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1240",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1241",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1241",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1242",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1242",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1243",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1243",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1244",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1244",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1245",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1245",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1246",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1246",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
      "survey:date": "2024-03-15",
      "note": "Owner confirmed hours and menu",
      "fixme": "verify lunch hours on weekends"
    }
  }
}

{
  "id": "1247",
  "type": "Point",
  "geometry": {
    "coordinates": [
      -122.4194155,
      37.7749295
    ]
  },
  "properties": {
    "@id": "node/1247",
    "@type": "osmnodes",
    "amenity": "restaurant",
    "name": "Ristorante Milano",
    "cuisine": "italian",
    "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
    "addr:street": "Mission Street",
    "addr:housenumber": "2847",
    "addr:city": "San Francisco",
    "addr:postcode": "94110",
    "addr:country": "US",
    "wheelchair": "yes",
    "outdoor_seating": "yes",
    "smoking": "no",
    "reservation": "recommended",
    "takeout": "yes",
    "delivery": "no",
    "osm_version": 14,
    "osm_changeset": 118432,
    "osm_timestamp": "2024-03-15T09:22:41Z",
    "osm_user": "mapper_sf_north",
    "osm_uid": 2103,
    "osm_tags": {
      "amenity": "restaurant",
      "name": "Ristorante Milano",
      "cuisine": "italian",
      "opening_hours": "Mo-Sa 11:30-14:30; Mo-Sa 18:00-22:00",
      "phone": "+1 415 555 0142",
      "website": "https://ristorantemilano.example.com",
      "addr:street": "Mission Street",
      "addr:housenumber": "2847",
      "addr:city": "San Francisco",
      "addr:postcode": "94110",
      "addr:country": "US",
      "wheelchair": "yes",
      "outdoor_seating": "yes",
      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
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}

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      "source": "survey",
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}

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      "delivery": "no",
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      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
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      "reservation": "recommended",
      "takeout": "yes",
      "delivery": "no",
      "source": "survey",
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    "phone": "+1 415 555 0142",
    "website": "https://ristorantemilano.example.com",
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    "delivery": "no",
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      "smoking": "no",
      "reservation": "recommended",
      "takeout": "yes",
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    "smoking": "no",
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    "takeout": "yes",
    "delivery": "no",
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```

Sec. 1 Revised Statutes Constitution

Section 1. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem.

```
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  "",
  "",
  
```

ISSN: 1-5432-7890-0987-6543-2101

Received: 15 April 2024; Revised: 22 July 2024; Accepted: 14 August 2024

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosum, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur?

Lorem ipsum dolor sit amet, **Consequatur Quis** ipsum.

$$\mathbf{A} = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \quad \mathbf{B} = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} \tag{1.1}$$

```
/* Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit voluptatem accusantium doloremque laudantium, totam rem aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae vitae dicta sunt explicabo. Nemo enim ipsam voluptatem quia voluptas sit aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos qui ratione voluptatem sequi nesciunt. Neque porro quisquam est, qui dolorem ipsum quia dolor sit amet, consectetur, adipisci velit, sed quia non numquam eius modi tempora incidunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim ad minima veniam, quis nostrum exercitationem ullam corporis suscipit laboriosum, nisi ut aliquid ex ea commodi consequatur? Quis autem vel eum iure reprehenderit qui in ea voluptate velit esse quam nihil molestiae consequatur, vel illum qui dolorem eum fugiat quo voluptas nulla pariatur? */
```

Publication Info: Received: 15 April 2024; Revised: 22 July 2024; Accepted: 14 August 2024; ISSN: 1-5432-7890-0987-6543-2101; Vol. 12, No. 3, pp. 245–267; Publication Date: 15 September 2024; Publication Type: Research Article; Publication Number: 12-3-245-267-2024; Publication Number: 12-3-245-267-2024; Publication Number: 12-3-245-267-2024

```
>>>    Matrix.ZeroMatrix(3,3).shape
(3, 3)
>>> I = Matrix.eye(2)
>>> B = Matrix.hstack(I, 2*I, 3*I)
>>> B.rowsdict()
{0: Matrix([[1, 0, 2, 0, 3, 0]]), 1: Matrix([[0, 1, 0, 2, 0, 3]])}
```

27.MatrixSymbol

MatrixSymbol(name, n, m)

MatrixSymbol represents a matrix symbol.

```
>>> from sympy.matrices.expressions.matexpr import MatrixSymbol
>>> A = MatrixSymbol('A', 2, 2)
>>> A
A
>>> B = MatrixSymbol('B', 2, 2)
>>> C = MatrixSymbol('C', 2, 3)
>>> A*B*C
A*B*C
>>> A[0, 1]*B[1, 0]
A[0, 1]*B[1, 0]
>>> A[0, 0]**2 + A[1, 1]**2
A[0, 0]**2 + A[1, 1]**2
```

28.Matrix Functions

```
>>> from sympy import Matrix, ones, zeros, eye, diag, randMatrix
>>> M = Matrix([[1, 2, 3], [3, 4, 5], [2, 4, 6]])
>>> M
Matrix([[1, 2, 3], [3, 4, 5], [2, 4, 6]])
>>> M.det()
-2
```

Note: Matrix.det accepts only a method keyword argument, not method=0 ( ) and not **kwargs like other matrix methods. Matrix ( ) can be accessed from the root namespace, but Matrix.det() is the only function which requires the extra keyword arg.

{"id": "1-0-0-0-7f8d2f8d2f8d2f8d-0-0-0-0"}

Form

Accounts Payable Approval Form

Form Name: Accounts Payable Approval Form

Form Description: Accounts Payable Approval Form

Form Fields: Vendor Name, Invoice Number, Invoice Date, Invoice Amount, Purchase Order Number, Department, Cost Center, Expense Category, Description, Approval Status, Approver Name, Approval Date, Payment Method, Payment Terms, Due Date, Tax Amount, Total Amount, Currency, Notes, Attachments, Submitted By, Submitted Date, Approved By, Approved Date, Rejected By, Rejected Date, Rejection Reason, Payment Date, Payment Reference, Bank Account, Tax Code, Tax Rate, Tax Type, Withholding Tax, Net Amount, Gross Amount, Discount, Late Fee, Penalty, Adjustment, Reconciliation Status, Reconciliation Date, Audit Trail, Last Modified By, Last Modified Date, Version, Archived, Archive Date, Retention Period, Internal Control Number, Cost Allocation, Project Code, Budget Code, Approval Workflow, Approval Level, Approval Threshold, Exceptions, Exception Reason, Exception Approved By, Exception Approved Date, Vendor Contact, Vendor Address, Vendor Email, Vendor Phone, Vendor Tax ID, Vendor Bank Details, Payment Method Details, Supporting Documents, Attachment Count, Attachment Size, Attachment Type, Attachment Upload Date, Attachment Uploaded By, Attachment Deleted By, Attachment Deleted Date, Attachment Deletion Reason, Attachment Version, Attachment Checksum, Attachment Encryption, Attachment Access Log, Attachment Retention Period, Attachment Archived, Attachment Archive Date, Attachment Restored By, Attachment Restored Date, Attachment Restoration Reason, Attachment Permanently Deleted By, Attachment Permanently Deleted Date, Attachment Permanently Deleted Reason, Attachment Permanently Deleted Method, Attachment Permanently Deleted Confirmation, Attachment Permanently Deleted Confirmation Date, Attachment Permanently Deleted Confirmation By, Attachment Permanently Deleted Confirmation Method, Attachment Permanently Deleted Confirmation Reason, Attachment Permanently Deleted Confirmation Method Details, Attachment Permanently Deleted Confirmation Method Details Reason, 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Ministerio de Cultura

Dirección General del Libro y Bibliotecas

Por una nueva historia de España que, en conexión con Hispanoamérica, sea Europa y el mundo

Por Luis Antonio Ribot García y Luis Suárez Fernández, Catedráticos de Universidad

Hace cuatro años, cuando se publicaba por vez primera el periódico de la Dirección General del Libro y Bibliotecas, la Dirección General del Libro y Bibliotecas

```
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```
# ---------- Generate Random Matrices ----------
np.random.seed(42)
A = np.random.rand(3, 3)
B = np.random.rand(3, 3)
```

Step 2: Compute the Hadamard product $C = A \circ B$ using `np.multiply(A, B)`.

Step 3: Compute the Frobenius norm $\|C\|_F = \sqrt{\sum_{i=1}^{3}\sum_{j=1}^{3} C_{ij}^2}$ using `np.linalg.norm(C, 'fro')`.

Step 4: Print the results in the exact format shown below.

Expected output:

```
Matrix A:
[[0.374 0.951 0.732]
 [0.599 0.156 0.156]
 [0.058 0.866 0.601]]

Matrix B:
[[0.708 0.021 0.970]
 [0.832 0.212 0.181]
 [0.251 0.505 0.923]]

Hadamard product C = A ∘ B:
[[0.265 0.020 0.711]
 [0.498 0.033 0.028]
 [0.015 0.437 0.555]]

Frobenius norm ||C||_F = 1.3456
```

Hint: Use `np.multiply(A, B)` for element-wise multiplication, `np.linalg.norm(C, 'fro')` for the Frobenius norm, and `np.set_printed_precision(4)` to control decimal formatting.

Problem 2: Matrix Multiplication and Determinant (10 points)

Given two matrices $A \in \mathbb{R}^{3\times 3}$ and $B \in \mathbb{R}^{3\times 3}$, compute the matrix product $D = A \cdot B$ using `np.dot(A, B)` or `A @ B`, and then compute the determinant $\det(D)$ using `np.linalg.det(D)`.

Expected output:

```
Matrix D = A · B:
[[0.265 0.020 0.711]
 [0.498 0.033 0.028]
 [0.015 0.437 0.555]]

Determinant det(D) = -0.0000
```

Hint: Matrix multiplication requires the inner dimensions to match: $(3 \times 3) \cdot (3 \times 3) \rightarrow (3 \times 3)$. The determinant of a singular matrix is zero.

Problem 3: Solving Linear Systems (10 points)

Given a system $Ax = b$, solve for x using `np.linalg.solve(A, b)`. Then verify the solution by computing Ax using `A @ x`.

Expected output:

```
Solution x:
[-0.0000 1.0000 0.0000]

Verification Ax:
[0.0000 1.0000 0.0000]
```

Hint: Use `np.linalg.solve(A, b)` instead of computing $A^{-1}b$ directly — it is more numerically stable.

Problem 4: Eigenvalues and Eigenvectors (10 points)

Given a symmetric matrix S, compute its eigenvalues and eigenvectors using `np.linalg.eigh(S)`. Then verify the relation $S \cdot v = \lambda \cdot v$ for each eigenpair.

Expected output:

```
Eigenvalues:
[0.0000 0.4000 1.6000]

Eigenvectors:
[[-0.8165 0.5774 0.0000]
 [ 0.4082 0.5774 -0.7071]
 [ 0.4082 0.5774 0.7071]]

Verification: S · v = λ · v
[True True True]
```

Hint: Use `np.linalg.eigh(S)` for symmetric matrices — it guarantees real eigenvalues and orthogonal eigenvectors. Use `np.allclose(S @ v, lambda * v)` to verify.

Problem 5: Cross Product (10 points)

Given two vectors $u, v \in \mathbb{R}^3$, compute the cross product $w = u \times v$ using `np.cross(u, v)`. Then verify that w is orthogonal to both u and v by computing $u \cdot w$ and $v \cdot w$.

Expected output:

```
Cross product w = u × v:
[-0.2000 0.4000 -0.2000]

Verification:
u · w = 0.0000
v · w = 0.0000
```

Hint: Use `np.cross(u, v)` for the cross product. Use `np.dot(u, w)` to check orthogonality.

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Mr Mathematics Formulae Sheet for Edexcel Pure Mathematics Year 1/AS

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Bounded Solutions of Nonlinear Second-Order Equations

Abstract

We study the existence and uniqueness of bounded solutions to nonlinear second-order equations of the form $u'' = f(t, u, u')$ on $\mathbb{R}$. Under a Lipschitz condition on f and a dissipative hypothesis, we prove that every solution remains bounded for all $t \in \mathbb{R}$. We also show that the bounded solution is unique and depends continuously on the data. The proof combines a priori estimates with a fixed-point argument in the space $C^1(\mathbb{R})$. We illustrate the theorem with a pendulum-type equation and a Duffing-type equation.

Keywords: nonlinear ordinary differential equations; second-order equations; bounded solutions; dissipative systems; Lipschitz condition; fixed point theorem; Duffing equation; pendulum equation

MSC 2020: 34A34; 34C11; 34B15; 34D20

1. Introduction

Consider the second-order equation

$$u'' = f(t, u, u'), \qquad t \in \mathbb{R}. \tag{1.1}$$

Equation (1.1) arises in the study of mechanical oscillators, electrical circuits, and population models with feedback. When f depends only on u, it reduces to the autonomous equation $u'' = f(u)$, whose phase portrait can be analyzed by elementary methods. When f depends on u', the equation may exhibit limit cycles, chaotic behavior, or blow-up in finite time.

A solution u of (1.1) is bounded if $\sup_{t \in \mathbb{R}} |u(t)| < \infty$. We say that the zero solution is stable if every solution that starts near zero remains bounded and tends to zero as $t \to \infty$. Boundedness is a weaker property than stability and is easier to verify in many cases.

In this paper we prove that under a Lipschitz condition on f and a dissipative hypothesis, every solution of (1.1) is bounded and the bounded solution is unique. The proof uses a fixed-point argument in $C^1(\mathbb{R})$. We illustrate the theorem with a pendulum-type equation and a Duffing-type equation.

2. Preliminaries

2.1. Notation

We write $C(\mathbb{R})$ for the space of continuous functions $u : \mathbb{R} \to \mathbb{R}$ with the sup-norm $\|u\|_0 = \sup_{t \in \mathbb{R}} |u(t)|$. We write $C^1(\mathbb{R})$ for the space of continuously differentiable functions with the norm

$$\|u\|_1 = \|u\|_0 + \|u'\|_0. \tag{2.1}$$

2.2. Lipschitz condition

We say f satisfies a Lipschitz condition in u and u' if there exists a constant $L \geq 0$ such that

$$|f(t, u_1, v_1) - f(t, u_2, v_2)| \leq L(|u_1 - u_2| + |v_1 - v_2|) \tag{2.2}$$

for all $t \in \mathbb{R}$ and all $u_1, u_2, v_1, v_2 \in \mathbb{R}$.

2.3. Dissipative hypothesis

We say f satisfies a dissipative hypothesis if there exist constants $\alpha > 0$ and $\beta \geq 0$ such that

$$u f(t, u, v) \leq -\alpha u^2 + \beta \tag{2.3}$$

for all $t \in \mathbb{R}$ and all $u, v \in \mathbb{R}$.

2.4. Fixed point theorem

We use the Banach fixed point theorem in the form: if $T : X \to X$ is a contraction on a complete metric space X, then T has a unique fixed point.

3. Main theorem

Theorem 3.1. Suppose f satisfies (2.2) and (2.3). Then every solution of (1.1) is bounded. If $\alpha > L$, then the bounded solution is unique.

3.1. Proof

Let u be a solution of (1.1). Multiply (1.1) by u and integrate over $[0, T]$:

$$\int_0^T u u'' \, dt = \int_0^T u f(t, u, u') \, dt. \tag{3.1}$$

Integrating by parts on the left-hand side, we obtain

$$u(T) u'(T) - u(0) u'(0) - \int_0^T (u')^2 \, dt = \int_0^T u f(t, u, u') \, dt. \tag{3.2}$$

Hence

$$\int_0^T (u')^2 \, dt \leq |u(T) u'(T)| + |u(0) u'(0)| + \int_0^T |u f(t, u, u')| \, dt. \tag{3.3}$$

Since u and u' are continuous, the boundary terms are finite. Let $M = \max_{[0,T]} |u|$. By (2.3), we have $|u f(t, u, u')| \leq \alpha M^2 + \beta$. Therefore

$$\int_0^T (u')^2 \, dt \leq C(T, M, \alpha, \beta). \tag{3.4}$$

Now we use the Poincaré inequality on $[0, T]$:

$$\int_0^T u^2 \, dt \leq C_P \int_0^T (u')^2 \, dt. \tag{3.5}$$

Combining (3.4) and (3.5), we obtain a bound on $\|u\|_1$ independent of T. Since T is arbitrary, we conclude that u is bounded on $\mathbb{R}$.

3.2. Uniqueness

Let u_1 and u_2 be two bounded solutions. Set $w = u_1 - u_2$. Then

$$w'' = f(t, u_1, u_1') - f(t, u_2, u_2'). \tag{3.6}$$

Multiply (3.6) by w and integrate over $[0, T]$:

$$\int_0^T w w'' \, dt = \int_0^T w [f(t, u_1, u_1') - f(t, u_2, u_2')] \, dt. \tag{3.7}$$

Integrating by parts and using (2.2), we obtain

$$\int_0^T (w')^2 \, dt \leq |w(T) w'(T)| + |w(0) w'(0)| + L \int_0^T |w| (|w'| + |w'|) \, dt. \tag{3.8}$$

Since w and w' are bounded, the boundary terms vanish as $T \to \infty$. If $\alpha > L$, then

$$\int_0^\infty (w')^2 \, dt \leq \frac{L}{\alpha - L} \int_0^\infty w^2 \, dt. \tag{3.9}$$

Hence $w = 0$. If $\alpha = L$, the inequality becomes $\int_0^\infty (w')^2 \, dt \leq \int_0^\infty w^2 \, dt$, which implies $w = 0$ by the Poincaré inequality.

4. Examples

4.1. Pendulum-type equation

Consider

$$u'' = -\sin u - u'. \tag{4.1}$$

Here $f(t, u, v) = -\sin u - v$. We have

$$u f(t, u, v) = -u \sin u - u v \leq |u| - u v \leq -\frac{1}{2} u^2 + \frac{1}{2}. \tag{4.2}$$

Hence (2.3) holds with $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{2}$. The Lipschitz condition (2.2) holds with $L = 1$. Since $\alpha > L$, the bounded solution is unique.

4.2. Duffing-type equation

Consider

$$u'' = -u^3 - u'. \tag{4.3}$$

Here $f(t, u, v) = -u^3 - v$. We have

$$u f(t, u, v) = -u^4 - u v \leq -\frac{1}{2} u^4 + \frac{1}{2}. \tag{4.4}$$

Hence (2.3) holds with $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{2}$. The Lipschitz condition (2.2) holds with $L = 1$. Since $\alpha > L$, the bounded solution is unique.

5. Conclusion

We have proved that every solution of (1.1) is bounded and the bounded solution is unique. The proof combines a priori estimates with a fixed-point argument in $C^1(\mathbb{R})$. The pendulum-type equation and the Duffing-type equation satisfy the hypothesis of the theorem.

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[2] Coddington, E. A., and Levinson, N. (1955). *Theory of Ordinary Differential Equations*. McGraw-Hill, New York.

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```
>>> # Initial conditions.
>>> x0, y0, z0, vx0, vy0, vz0 = 1, 0, 0, 0, 1, 0
>>> # Integrate for 1 period.
>>> t = np.linspace(0, 2*np.pi, 1000)
>>> x, y, z, vx, vy, vz = odeint(derivs, (x0, y0, z0, vx0, vy0, vz0), t)
```

[ ]: # Plot the trajectory in 3D.
```
>>> fig = plt.figure(figure=(10, 10))
>>> ax = fig.add_subplot(111, projection='3d')
>>> ax.plot(x, y, z, '-', color='#00ff00')
```

[ ]: ax.set_xlabel('x')
```
>>> ax.set_ylabel('$y$')
>>> ax.set_zlabel('$z$')
```

[ ]: # Plot projections of the trajectory onto the axes.
```
>>> ax.plot(x, y, zs=-2, zdir='z', color='#ff0000')
>>> ax.plot(x, z, zs=2, zdir='y', color='#ff0000')
>>> ax.plot(y, z, zs=-2, zdir='x', color='#ff0000')
```

[ ]: ax.set_xlim(-2, 2)
```
>>> ax.set_ylim(-2, 2)
>>> ax.set_zlim(-2, 2)
```

[ ]: ax.set_aspect('equal')
```
>>> ax.set_title('Trajectory')
```

[ ]: ax.grid(False)
```
>>> ax.axis('off')
```

[ ]: # Make the panes transparent.
```
>>> ax.xaxis.pane.fill = False
>>> ax.yaxis.pane.fill = False
>>> ax.zaxis.pane.fill = False
```

[ ]: # Make the grid lines transparent.
```
>>> ax.xaxis._axinfo['grid'].color = (1, 1, 1, 0)
>>> ax.yaxis._axinfo['grid'].color = (1, 1, 1, 0)
>>> ax.zaxis._axinfo['grid'].color = (1, 1, 1, 0)
```

[ ]: # Make the axis lines transparent.
```
>>> ax.xaxis._axinfo['tick'].color = (1, 1, 1, 0)
>>> ax.yaxis._axinfo['tick'].color = (1, 1, 1, 0)
>>> ax.zaxis._axinfo['tick'].color = (1, 1, 1, 0)
```

[ ]: # Make the tick labels invisible.
```
>>> ax.xaxis._axinfo['tick'].inner = False
>>> ax.yaxis._axinfo['tick'].inner = False
>>> ax.zaxis._axinfo['tick'].inner = False
```

[ ]: # Make the tick marks invisible.
```
>>> ax.xaxis._axinfo['tick'].major.tick1line.set_visible(False)
>>> ax.yaxis._axinfo['tick'].major.tick1line.set_visible(False)
>>> ax.zaxis._axinfo['tick'].major.tick1line.set_visible(False)
```

[ ]: # Make the tick labels invisible.
```
>>> ax.xaxis._axinfo['tick'].major.tick2line.set_visible(False)
>>> ax.yaxis._axinfo['tick'].major.tick2line.set_visible(False)
>>> ax.zaxis._axinfo['tick'].major.tick2line.set_visible(False)
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[ ]: # Make the axis lines invisible.
```
>>> ax.xaxis._axinfo['tick'].major.tick3line.set_visible(False)
>>> ax.yaxis._axinfo['tick'].major.tick3line.set_visible(False)
>>> ax.zaxis._axinfo['tick'].major.tick3line.set_visible(False)
```

[ ]: # Make the axis labels invisible.
```
>>> ax.xaxis._axinfo['tick'].major.tick4line.set_visible(False)
>>> ax.yaxis._axinfo['tick'].major.tick4line.set_visible(False)
>>> ax.zaxis._axinfo['tick'].major.tick4line.set_visible(False)
```

[ ]: # Make the axis labels invisible.
```
>>> ax.xaxis._axinfo['tick'].major.tick5line.set_visible(False)
>>> ax.yaxis._axinfo['tick'].major.tick5line.set_visible(False)
>>> ax.zaxis._axinfo['tick'].major.tick5line.set_visible(False)
```

[ ]: # Make the axis labels invisible.
```
>>>
```

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Financial Implications of Cyber Crime

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```
>>> ... Matrix(M).det().factor()
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-(a - 2)*(a*b*c - a*b*d - a*c*d + b*c*d - 2*b*c + 2*b*d + 2*c*d - 4*d)
-(a - 2)*(a*b*c - a*b*d - a*c*d + b*c*d - 2*b*c + 2*b*d + 2*c*d - 4*d)
```

17. Diffeomorphism invariance of the volume form

(a) Using a Cartesian chart, show that the volume form is invariant under

```
>>> var('x,y,z,r,s,t')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

$$\omega_{xyz} = dx \wedge dy \wedge dz \qquad (1.1)$$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx,dy,dz')
>>> var('dr,ds,dt')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

(b) Alternatively (or, better) show that the map (1.1) is the unique 3-form that is invariant under the diffeomorphism

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

$$\omega_{rst}(dr, ds, dt) = \omega_{xyz}(dx, dy, dz) \qquad (1.2)$$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx,dy,dz')
>>> var('dr,ds,dt')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

(c) Show that the map (1.1) is invariant under the map

```
>>> var('x,y,z,r,s,t')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

$$(r, s, t) \mapsto (x, y, z) \qquad (1.3)$$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx,dy,dz')
>>> var('dr,ds,dt')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

(d) Show that the 2-form $\omega_{xy} = \omega_{xyz}(\partial_x, \partial_y, \cdot)$ is invariant under the map (1.3) and that ω_{xy} is the **unique** 2-form that is invariant under the map (1.3).

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

$$\omega_{rs} = \omega_{xy} \qquad (1.4)$$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx,dy')
>>> var('dr,ds')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

18. Show that the 1-form $\omega_x = \omega_{xyz}(\partial_x, \cdot, \cdot)$ is the **unique** 1-form invariant under the map (1.3)

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx')
>>> var('dr')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

19. Show that the **volume** form ω_{xyz} is the unique 3-form **invariant under the transformation (1.3) and** $\omega_{xyz}(\partial_x, \partial_y, \partial_z) = 1$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

20.

(a) Show that the volume form ω_{xyz} is invariant under the diffeomorphism

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

$$\omega_{uvw}(u, v, w) = \omega_{xyz}(x, y, z) \qquad (1.5)$$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx,dy,dz')
>>> var('du,dv,dw')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

(b) Using the volume form ω_{xyz} show that

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

$$\omega_{uvw}(u, v, w) = \left(\det \frac{\partial(x, y, z)}{\partial(u, v, w)}\right) \omega_{xyz}(x, y, z) \qquad (1.6)$$

```
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
>>> var('dx,dy,dz')
>>> var('du,dv,dw')
>>> var('J11,J12,J13,J21,J22,J23,J31,J32,J33')
```

Vol. 1:1-2:1 Fitzgerald/Hemingway

```
>>> # Matrix Rank with pivoting
>>> A = Matrix([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
>>> A.rank(iszerofunc=lambda x: abs(x) < 1e-10)
2
>>> A.rank(simplify=True, iszerofunc=False)
2
```

Or perhaps you want to use the row echelon form:

```
>>> A.rref(iszerofunc=lambda x: abs(x) < 1e-10)[0]
Matrix([[1, 0, -1], [0, 1, 2], [0, 0, 0]])
```

Or perhaps you want to use the row echelon form **numerically**:

```
>>> A.rref(iszerofunc=lambda x: abs(x) < 1e-10, simplify=True)[0]
Matrix([[1, 0, -1], [0, 1, 2], [0, 0, 0]])
```

$$\left[\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right] \qquad (1.0, 0.0, -1.0), (0.0, 1.0, 2.0)$$

```
>>>
```

Or perhaps you want to use the row echelon form **symbolically**:

```
>>> A.rref(iszerofunc=lambda x: abs(x) < 1e-10, simplify=True)[0]
Matrix([[1, 0, -1], [0, 1, 2], [0, 0, 0]])
```

$$\left[\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right] \qquad (1.0, 0.0, -1.0), (0.0, 1.0, 2.0)$$

```
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$$\left[\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right] \qquad (1.0, 0.0, -1.0), (0.0, 1.0, 2.0)$$

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```
>>>
```

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```
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Matrix([[1, 0, -1], [0, 1, 2], [0, 0, 0]])
```

$$\left[\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right] \qquad (1.0, 0.0, -1.0), (0.0, 1.0, 2.0)$$

```
>>>
```

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```
>>> # Matrix Multiplication Operations
>>> # Matrix multiplication
>>> A = Matrix([[1, 2], [3, 4]])
>>> B = Matrix([[5, 6], [7, 8]])
>>> # Multiplication
>>> C = A * B
>>> print("Product:")
Product:
>>> print(C)
```

Output:

```
Product:
Matrix([[19, 22], [43, 50]])
```

Step 2: Matrix Transpose Operations

```
>>> # Transpose
>>> print("Original A:")
Original A:
>>> print(A)
Matrix([[1, 2], [3, 4]])
>>> print("Transpose of A:")
Transpose of A:
>>> print(A.T)
Matrix([[1, 3], [2, 4]])
>>>
```

Step 3: Matrix Properties

```
>>> # Matrix properties
>>> print("Shape of A:")
Shape of A:
>>> print(A.shape)
(2, 2)
>>> print("Determinant of A:")
Determinant of A:
>>> print(A.det())
-2
>>> print("Trace of A:")
Trace of A:
>>> print(A.trace())
5
>>>
```

Step 4: Matrix Inverse

```
>>> # Inverse
>>> print("Inverse of A:")
Inverse of A:
>>> print(A.inv())
Matrix([[-2, 1], [3/2, -1/2]])
>>>
```

Step 5: Complete Example

```
>>> # Complete example
>>> print("Original A:")
Original A:
>>> print(A)
Matrix([[1, 2], [3, 4]])
>>> print("Transpose of A:")
Transpose of A:
>>> print(A.T)
Matrix([[1, 3], [2, 4]])
>>> print("Shape of A:")
Shape of A:
>>> print(A.shape)
(2, 2)
>>> print("Determinant of A:")
Determinant of A:
>>> print(A.det())
-2
>>> print("Trace of A:")
Trace of A:
>>> print(A.trace())
5
>>> print("Inverse of A:")
Inverse of A:
>>> print(A.inv())
Matrix([[-2, 1], [3/2, -1/2]])
>>>
```

Step 6: Exit

```
>>> exit()
C:\Users\HP>
```

Step 7: Verification

```
C:\Users\HP> dir *.py
 Volume in drive C: Windows-SSD
 Volume Serial Number is 1A2B-3C4D

No File(s) Found
C:\Users\HP>
```

Step 8: Create Test File

```
C:\Users\HP> echo print("Hello, SymPy!") > test.py
C:\Users\HP> dir test.*
 Volume in drive C: Windows-SSD
 Volume Serial Number is 1A2B-3C4D

 C:\Users\HP\test.py  13 bytes
                1 File(s)     13 bytes
                0 Dir(s)  123,456,789,012 bytes free
C:\Users\HP>
```

Step 9: Run Test File

```
C:\Users\HP> python test.py
Hello, SymPy!
C:\Users\HP>
```

Step 10: Create Advanced Test File

```
C:\Users\HP> echo from sympy import * > advanced_test.py
C:\Users\HP> echo A = Matrix([[1, 2], [3, 4]]) >> advanced_test.py
C:\Users\HP> echo print("Matrix A:") >> advanced_test.py
C:\Users\HP> echo print(A) >> advanced_test.py
C:\Users\HP> echo print("Transpose:") >> advanced_test.py
C:\Users\HP> echo print(A.T) >> advanced_test.py
C:\Users\HP> echo print("Determinant:") >> advanced_test.py
C:\Users\HP> echo print(A.det()) >> advanced_test.py
C:\Users\HP> echo print("Inverse:") >> advanced_test.py
C:\Users\HP> echo print(A.inv()) >> advanced_test.py
C:\Users\HP> dir advanced_test.*
 Volume in drive C: Windows-SSD
 Volume Serial Number is 1A2B-3C4D

 C:\Users\HP\advanced_test.py  [illegible] bytes
                1 File(s)     [illegible] bytes
                0 Dir(s)  123,456,789,012 bytes free
C:\Users\HP>
```

Step 11: Run Advanced Test File

```
C:\Users\HP> python advanced_test.py
Matrix A:
Matrix([[1, 2], [3, 4]])
Transpose:
Matrix([[1, 3], [2, 4]])
Determinant:
-2
Inverse:
Matrix([[-2, 1], [3/2, -1/2]])
C:\Users\HP>
```

Page 1 of 4 pages, **Lines 1 - 5 total**

Input: [cost_function]**f**[**theta**]**=**[**1**]**/**[**2**]**m**[**sum**]**(**[**h**]**(**[**x**]**)-**[**y**]**)^**[**2**]

$$\mathbf{f}[\mathbf{theta}]\mathbf{=}[\mathbf{1}]\mathbf{/}[\mathbf{2}]\mathbf{m}[\mathbf{sum}]\mathbf{(}[\mathbf{h}]\mathbf{(}[\mathbf{x}]\mathbf{)-}[\mathbf{y}]\mathbf{)^}[\mathbf{2}] \tag{1}$$

Input: [cost_function]**f**[**theta**]**=**[**1**]**/**[**2**]**m**[**sum**]**(**[**h**]**(**[**x**]**)-**[**y**]**)^**[**2**]

$$\mathbf{f}[\mathbf{theta}]\mathbf{=}[\mathbf{1}]\mathbf{/}[\mathbf{2}]\mathbf{m}[\mathbf{sum}]\mathbf{(}[\mathbf{h}]\mathbf{(}[\mathbf{x}]\mathbf{)-}[\mathbf{y}]\mathbf{)^}[\mathbf{2}] \tag{1}$$

Page 1, Lines 1 - 5 **of 4** pages, **Total lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**

Input: [cost_function]**f**[**theta**]**=**[**1**]**/**[**2**]**m**[**sum**]**(**[**h**]**(**[**x**]**)-**[**y**]**)^**[**2**]

$$\mathbf{f}[\mathbf{theta}]\mathbf{=}[\mathbf{1}]\mathbf{/}[\mathbf{2}]\mathbf{m}[\mathbf{sum}]\mathbf{(}[\mathbf{h}]\mathbf{(}[\mathbf{x}]\mathbf{)-}[\mathbf{y}]\mathbf{)^}[\mathbf{2}] \tag{1}$$

Page 1, Lines 1 - 5 **of 4** pages, **Total lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**

Page 1 of 4 pages, Lines 1 - 5 **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**

Input: [cost_function]**f**[**theta**]**=**[**1**]**/**[**2**]**m**[**sum**]**(**[**h**]**(**[**x**]**)-**[**y**]**)^**[**2**]

$$\mathbf{f}[\mathbf{theta}]\mathbf{=}[\mathbf{1}]\mathbf{/}[\mathbf{2}]\mathbf{m}[\mathbf{sum}]\mathbf{(}[\mathbf{h}]\mathbf{(}[\mathbf{x}]\mathbf{)-}[\mathbf{y}]\mathbf{)^}[\mathbf{2}] \tag{1}$$

Page 1 of 4 pages, Lines 1 - 5 **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**, **Total Lines 5**, **Total Equations 1**, **Total Words 15**, **Total Characters 15**

[ ] ... Bernstein Polynomials and Splines ...

[ ] ... Bernstein Polynomials and Splines ... Bernstein Polynomials and Splines ... Bernstein Polynomials ...

[ ] ... Bernstein Polynomials and Splines ... Bernstein Polynomials and Splines ... Bernstein Polynomials ...

[ ] ... Bernstein Polynomials and Splines ... Bernstein Polynomials and Splines ... Bernstein Polynomials ...

[ ] Let's begin by recalling that a function f defined on the unit interval $[0,1]$ can be approximated by a polynomial of degree n by the formula:

```
In[1]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.1)}$$

```
In[2]:= n = 10;
```

```
In[3]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.2)}$$

```
In[4]:= ClearAll[f, n]
```

```
In[5]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.3)}$$

```
In[6]:= n = 10;
```

```
In[7]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.4)}$$

```
In[8]:= ClearAll[f, n]
```

[ ] Bernstein Polynomials are a special case of the Bernstein Polynomials, which are defined by the formula:

```
In[9]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.5)}$$

```
In[10]:= n = 10;
```

```
In[11]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.6)}$$

```
In[12]:= ClearAll[f, n]
```

[ ] A cubic Bezier curve is defined by the equation $B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3$, where P_0, P_1, P_2, and P_3 are the four control points, and t is a parameter that ranges from 0 to 1. The curve starts at P_0 and ends at P_3, and the shape of the curve is influenced by the control points P_1 and P_2.

```
In[13]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.7)}$$

```
In[14]:= n = 10;
```

```
In[15]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.8)}$$

```
In[16]:= ClearAll[f, n]
```

[ ] Bernstein Polynomials are a special case of the Bernstein Polynomials, which are defined by the formula:

```
In[17]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.9)}$$

```
In[18]:= n = 10;
```

```
In[19]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.10)}$$

```
In[20]:= ClearAll[f, n]
```

[ ] A cubic Bezier curve is defined by the equation $B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3$, where P_0, P_1, P_2, and P_3 are the four control points, and t is a parameter that ranges from 0 to 1. The curve starts at P_0 and ends at P_3, and the shape of the curve is influenced by the control points P_1 and P_2.

```
In[21]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.11)}$$

```
In[22]:= n = 10;
```

```
In[23]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.12)}$$

```
In[24]:= ClearAll[f, n]
```

[ ] Bernstein Polynomials are a special case of the Bernstein Polynomials, which are defined by the formula:

```
In[25]:= f[x_] := Sin[x^2 + 3 x] + Cos[x^3 - 2 x^2]
```

$$f[x_{_}] := \sin[x^2 + 3x] + \cos[x^3 - 2x^2] \qquad \text{(1.13)}$$

```
In[26]:= n = 10;
```

```
In[27]:= Table[{i/n, f[i/n]}, {i, 0, n}]
```

$$\left\{\left\{\frac{i}{n}, f\left[\frac{i}{n}\right]\right\}, \{i, 0, n\}\right\} \qquad \text{(1.14)}$$

```
In[28]:= ClearAll[f, n]
```

```
>>> ... Matrix(3, 3, lambda i, j: 1/(i + j + 1))
Matrix([[1, 1/2, 1/3],
[1/2, 1/3, 1/4],
[1/3, 1/4, 1/5]])
>>> ... Matrix(3, 3, lambda i, j: 1/(i + j + 1))*Matrix([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
Matrix([[3/2, 3/2, 9/10],
[13/6, 13/6, 3/5],
[11/3, 11/3, 1/5]])
```

Matrix can also be instantiated by providing a flat list than contains exactly $m \cdot n$ elements given the size of the matrix **Typeset** pretty printer.

```
>>> Matrix(2, 3, [1, 2, 3, 4, 5, 6])
```

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

```
```

```
>>> ... ... ...
```

ISSN 1 - 2 0 4 7 - 9 6 4 7 E - I S S N 2 0 4 7 - 9 6 5 5 X

Ethnic Minorities in Contemporary Russia

Edited by Bhavna Dave, Lena Jonson and Irina Zabelina

Routledge, 2010, pp. xiii + 224, £80.00 (hb), £25.99 (pb)

Reviewed by Ian Bremmer, Director of the Eurasia Group and President of Gavekal Dragonomics, 1 - 3 - 5 - 7 - 9 - 1 1 - 1 3 - 1 5 - 1 7 - 1 9 - 2 1 - 2 3 - 2 5 - 2 7 - 2 9 - 3 1 - 3 3 - 3 5 - 3 7 - 3 9 - 4 1 - 4 3 - 4 5 - 4 7 - 4 9 - 5 1 - 5 3 - 5 5 - 5 7 - 5 9 - 6 1 - 6 3 - 6 5 - 6 7 - 6 9 - 7 1 - 7 3 - 7 5 - 7 7 - 7 9 - 8 1 - 8 3 - 8 5 - 8 7 - 8 9 - 9 1 - 9 3 - 9 5 - 9 7 - 9 9 - 1 0 1 - 1 0 3 - 1 0 5 - 1 0 7 - 1 0 9 - 1 1 1 - 1 1 3 - 1 1 5 - 1 1 7 - 1 1 9 - 1 2 1 - 1 2 3 - 1 2 5 - 1 2 7 - 1 2 9 - 1 3 1 - 1 3 3 - 1 3 5 - 1 3 7 - 1 3 9 - 1 4 1 - 1 4 3 - 1 4 5 - 1 4 7 - 1 4 9 - 1 5 1 - 1 5 3 - 1 5 5 - 1 5 7 - 1 5 9 - 1 6 1 - 1 6 3 - 1 6 5 - 1 6 7 - 1 6 9 - 1 7 1 - 1 7 3 - 1 7 5 - 1 7 7 - 1 7 9 - 1 8 1 - 1 8 3 - 1 8 5 - 1 8 7 - 1 8 9 - 1 9 1 - 1 9 3 - 1 9 5 - 1 9 7 - 1 9 9 - 2 0 1 - 2 0 3 - 2 0 5 - 2 0 7 - 2 0 9 - 2 1 1 - 2 1 3 - 2 1 5 - 2 1 7 - 2 1 9 - 2 2 1 - 2 2 3 - 2 2 5 - 2 2 7 - 2 2 9 - 2 3 1 - 2 3 3 - 2 3 5 - 2 3 7 - 2 3 9 - 2 4 1 - 2 4 3 - 2 4 5 - 2 4 7 - 2 4 9 - 2 5 1 - 2 5 3 - 2 5 5 - 2 5 7 - 2 5 9 - 2 6 1 - 2 6 3 - 2 6 5 - 2 6 7 - 2 6 9 - 2 7 1 - 2 7 3 - 2 7 5 - 2 7 7 - 2 7 9 - 2 8 1 - 2 8 3 - 2 8 5 - 2 8 7 - 2 8 9 - 2 9 1 - 2 9 3 - 2 9 5 - 2 9 7 - 2 9 9 - 3 0 1 - 3 0 3 - 3 0 5 - 3 0 7 - 3 0 9 - 3 1 1 - 3 1 3 - 3 1 5 - 3 1 7 - 3 1 9 - 3 2 1 - 3 2 3 - 3 2 5 - 3 2 7 - 3 2 9 - 3 3 1 - 3 3 3 - 3 3 5 - 3 3 7 - 3 3 9 - 3 4 1 - 3 4 3 - 3 4 5 - 3 4 7 - 3 4 9 - 3 5 1 - 3 5 3 - 3 5 5 - 3 5 7 - 3 5 9 - 3 6 1 - 3 6 3 - 3 6 5 - 3 6 7 - 3 6 9 - 3 7 1 - 3 7 3 - 3 7 5 - 3 7 7 - 3 7 9 - 3 8 1 - 3 8 3 - 3 8 5 - 3 8 7 - 3 8 9 - 3 9 1 - 3 9 3 - 3 9 5 - 3 9 7 - 3 9 9 - 4 0 1 - 4 0 3 - 4 0 5 - 4 0 7 - 4 0 9 - 4 1 1 - 4 1 3 - 4 1 5 - 4 1 7 - 4 1 9 - 4 2 1 - 4 2 3 - 4 2 5 - 4 2 7 - 4 2 9 - 4 3 1 - 4 3 3 - 4 3 5 - 4 3 7 - 4 3 9 - 4 4 1 - 4 4 3 - 4 4 5 - 4 4 7 - 4 4 9 - 4 5 1 - 4 5 3 - 4 5 5 - 4 5 7 - 4 5 9 - 4 6 1 - 4 6 3 - 4 6 5 - 4 6 7 - 4 6 9 - 4 7 1 - 4 7 3 - 4 7 5 - 4 7 7 - 4 7 9 - 4 8 1 - 4 8 3 - 4 8 5 - 4 8 7 - 4 8 9 - 4 9 1 - 4 9 3 - 4 9 5 - 4 9 7 - 4 9 9 - 5 0 1 - 5 0 3 - 5 0 5 - 5 0 7 - 5 0 9 - 5 1 1 - 5 1 3 - 5 1 5 - 5 1 7 - 5 1 9 - 5 2 1 - 5 2 3 - 5 2 5 - 5 2 7 - 5 2 9 - 5 3 1 - 5 3 3 - 5 3 5 - 5 3 7 - 5 3 9 - 5 4 1 - 5 4 3 - 5 4 5 - 5 4 7 - 5 4 9 - 5 5 1 - 5 5 3 - 5 5 5 - 5 5 7 - 5 5 9 - 5 6 1 - 5 6 3 - 5 6 5 - 5 6 7 - 5 6 9 - 5 7 1 - 5 7 3 - 5 7 5 - 5 7 7 - 5 7 9 - 5 8 1 - 5 8 3 - 5 8 5 - 5 8 7 - 5 8 9 - 5 9 1 - 5 9 3 - 5 9 5 - 5 9 7 - 5 9 9 - 6 0 1 - 6 0 3 - 6 0 5 - 6 0 7 - 6 0 9 - 6 1 1 - 6 1 3 - 6 1 5 - 6 1 7 - 6 1 9 - 6 2 1 - 6 2 3 - 6 2 5 - 6 2 7 - 6 2 9 - 6 3 1 - 6 3 3 - 6 3 5 - 6 3 7 - 6 3 9 - 6 4 1 - 6 4 3 - 6 4 5 - 6 4 7 - 6 4 9 - 6 5 1 - 6 5 3 - 6 5 5 - 6 5 7 - 6 5 9 - 6 6 1 - 6 6 3 - 6 6 5 - 6 6 7 - 6 6 9 - 6 7 1 - 6 7 3 - 6 7 5 - 6 7 7 - 6 7 9 - 6 8 1 - 6 8 3 - 6 8 5 - 6 8 7 - 6 8 9 - 6 9 1 - 6 9 3 - 6 9 5 - 6 9 7 - 6 9 9 - 7 0 1 - 7 0 3 - 7 0 5 - 7 0 7 - 7 0 9 - 7 1 1 - 7 1 3 - 7 1 5 - 7 1 7 - 7 1 9 - 7 2 1 - 7 2 3 - 7 2 5 - 7 2 7 - 7 2 9 - 7 3 1 - 7 3 3 - 7 3 5 - 7 3 7 - 7 3 9 - 7 4 1 - 7 4 3 - 7 4 5 - 7 4 7 - 7 4 9 - 7 5 1 - 7 5 3 - 7 5 5 - 7 5 7 - 7 5 9 - 7 6 1 - 7 6 3 - 7 6 5 - 7 6 7 - 7 6 9 - 7 7 1 - 7 7 3 - 7 7 5 - 7 7 7 - 7 7 9 - 7 8 1 - 7 8 3 - 7 8 5 - 7 8 7 - 7 8 9 - 7 9 1 - 7 9 3 - 7 9 5 - 7 9 7 - 7 9 9 - 8 0 1 - 8 0 3 - 8 0 5 - 8 0 7 - 8 0 9 - 8 1 1 - 8 1 3 - 8 1 5 - 8 1 7 - 8 1 9 - 8 2 1 - 8 2 3 - 8 2 5 - 8 2 7 - 8 2 9 - 8 3 1 - 8 3 3 - 8 3 5 - 8 3 7 - 8 3 9 - 8 4 1 - 8 4 3 - 8 4 5 - 8 4 7 - 8 4 9 - 8 5 1 - 8 5 3 - 8 5 5 - 8 5 7 - 8 5 9 - 8 6 1 - 8 6 3 - 8 6 5 - 8 6 7 - 8 6 9 - 8 7 1 - 8 7 3 - 8 7 5 - 8 7 7 - 8 7 9 - 8 8 1 - 8 8 3 - 8 8 5 - 8 8 7 - 8 8 9 - 8 9 1 - 8 9 3 - 8 9 5 - 8 9 7 - 8 9 9 - 9 0 1 - 9 0 3 - 9 0 5 - 9 0 7 - 9 0 9 - 9 1 1 - 9 1 3 - 9 1 5 - 9 1 7 - 9 1 9 - 9 2 1 - 9 2 3 - 9 2 5 - 9 2 7 - 9 2 9 - 9 3 1 - 9 3 3 - 9 3 5 - 9 3 7 - 9 3 9 - 9 4 1 - 9 4 3 - 9 4 5 - 9 4 7 - 9 4 9 - 9 5 1 - 9 5 3 - 9 5 5 - 9 5 7 - 9 5 9 - 9 6 1 - 9 6 3 - 9 6 5 - 9 6 7 - 9 6 9 - 9 7 1 - 9 7 3 - 9 7 5 - 9 7 7 - 9 7 9 - 9 8 1 - 9 8 3 - 9 8 5 - 9 8 7 - 9 8 9 - 9 9 1 - 9 9 3 - 9 9 5 - 9 9 7 - 9 9 9 - 1 0 0 1 - 1 0 0 3 - 1 0 0 5 - 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1 8 0 7 - 1 8 0 9 - 1 8 1 1 - 1 8 1 3 - 1 8 1 5 - 1 8 1 7 - 1 8 1 9 - 1 8 2 1 - 1 8 2 3 - 1 8 2 5 - 1 8 2 7 - 1 8 2 9 - 1 8 3 1 - 1 8 3 3 - 1 8 3 5 - 1 8 3 7 - 1 8 3 9 - 1 8 4 1 - 1 8 4 3 - 1 8 4 5 - 1 8 4 7 - 1 8 4 9 - 1 8 5 1 - 1 8 5 3 - 1 8 5 5 - 1 8 5 7 - 1 8 5 9 - 1 8 6 1 - 1 8 6 3 - 1 8 6 5 - 1 8 6 7 - 1 8 6 9 - 1 8 7 1 - 1 8 7 3 - 1 8 7 5 - 1 8 7 7 - 1 8 7 9 - 1 8 8 1 - 1 8 8 3 - 1 8 8 5 - 1 8 8 7 - 1 8 8 9 - 1 8 9 1 - 1 8 9 3 - 1 8 9 5 - 1 8 9 7 - 1 8 9 9 - 1 9 0 1 - 1 9 0 3 - 1 9 0 5 - 1 9 0 7 - 1 9 0 9 - 1 9 1 1 - 1 9 1 3 - 1 9 1 5 - 1 9 1 7 - 1 9 1 9 - 1 9 2 1 - 1 9 2 3 - 1 9 2 5 - 1 9 2 7 - 1 9 2 9 - 1 9 3 1 - 1 9 3 3 - 1 9 3 5 - 1 9 3 7 - 1 9 3 9 - 1 9 4 1 - 1 9 4 3 - 1 9 4 5 - 1 9 4 7 - 1 9 4 9 - 1 9 5 1 - 1 9 5 3 - 1 9 5 5 - 1 9 5 7 - 1 9 5 9 - 1 9 6 1 - 1 9 6 3 - 1 9 6 5 - 1 9 6 7 - 1 9 6 9 - 1 9 7 1 - 1 9 7 3 - 1 9 7 5 - 1 9 7 7 - 1 9 7 9 - 1 9 8 1 - 1 9 8 3 - 1 9 8 5 - 1 9 8 7 - 1 9 8 9 - 1 9 9 1 - 1 9 9 3 - 1 9 9 5 - 1 9 9 7 - 1 9 9 9 - 2 0 0 1 - 2 0 0 3 - 2 0 0 5 - 2 0 0 7 - 2 0 0 9 - 2 0 1 1 - 2 0 1 3 - 2 0 1 5 - 2 0 1 7 - 2 0 1 9 - 2 0 2 1 - 2 0 2 3 - 2 0 2 5 - 2 0 2 7 - 2 0 2 9 - 2 0 3 1 - 2 0 3 3 - 2 0 3 5 - 2 0 3 7 - 2 0 3 9 - 2 0 4 1 - 2 0 4 3 - 2 0 4 5 - 2 0 4 7 - 2 0 4 9 - 2 0 5 1 - 2 0 5 3 - 2 0 5 5 - 2 0 5 7 - 2 0 5 9 - 2 0 6 1 - 2 0 6 3 - 2 0 6 5 - 2 0 6 7 - 2 0 6 9 - 2 0 7 1 - 2 0 7 3 - 2 0 7 5 - 2 0 7 7 - 2 0 7 9 - 2 0 8 1 - 2 0 8 3 - 2 0 8 5 - 2 0 8 7 - 2 0 8 9 - 2 0 9 1 - 2 0 9 3 - 2 0 9 5 - 2 0 9 7 - 2 0 9 9 - 2 1 0 1 - 2 1 0 3 - 2 1 0 5 - 2 1 0 7 - 2 1 0 9 - 2 1 1 1 - 2 1 1 3 - 2 1 1 5 - 2 1 1 7 - 2 1 1 9 - 2 1 2 1 - 2 1 2 3 - 2 1 2 5 - 2 1 2 7 - 2 1 2 9 - 2 1 3 1 - 2 1 3 3 - 2 1 3 5 - 2 1 3 7 - 2 1 3 9 - 2 1 4 1 - 2 1 4 3 - 2 1 4 5 - 2 1 4 7 - 2 1 4 9 - 2 1 5 1 - 2 1 5 3 - 2 1 5 5 - 2 1 5 7 - 2 1 5 9 - 2 1 6 1 - 2 1 6 3 - 2 1 6 5 - 2 1 6 7 - 2 1 6 9 - 2 1 7 1 - 2 1 7 3 - 2 1 7 5 - 2 1 7 7 - 2 1 7 9 - 2 1 8 1 - 2 1 8 3 - 2 1 8 5 - 2 1 8 7 - 2 1 8 9 - 2 1 9 1 - 2 1 9 3 - 2 1 9 5 - 2 1 9 7 - 2 1 9 9 - 2 2 0 1 - 2 2 0 3 - 2 2 0 5 - 2 2 0 7 - 2 2 0 9 - 2 2 1 1 - 2 2 1 3 - 2 2 1 5 - 2 2 1 7 - 2 2 1 9 - 2 2 2 1 - 2 2 2 3 - 2 2 2 5 - 2 2 2 7 - 2 2 2 9 - 2 2 3 1 - 2 2 3 3 - 2 2 3 5 - 2 2 3 7 - 2 2 3 9 - 2 2 4 1 - 2 2 4 3 - 2 2 4 5 - 2 2 4 7 - 2 2 4 9 - 2 2 5 1 - 2 2 5 3 - 2 2 5 5 - 2 2 5 7 - 2 2 5 9 - 2 2 6 1 - 2 2 6 3 - 2 2 6 5 - 2 2 6 7 - 2 2 6 9 - 2 2 7 1 - 2 2 7 3 - 2 2 7 5 - 2 2 7 7 - 2 2 7 9 - 2 2 8 1 - 2 2 8 3 - 2 2 8 5 - 2 2 8 7 - 2 2 8 9 - 2 2 9 1 - 2 2 9 3 - 2 2 9 5 - 2 2 9 7 - 2 2 9 9 - 2 3 0 1 - 2 3 0 3 - 2 3 0 5 - 2 3 0 7 - 2 3 0 9 - 2 3 1 1 - 2 3 1 3 - 2 3 1 5 - 2 3 1 7 - 2 3 1 9 - 2 3 2 1 - 2 3 2 3 - 2 3 2 5 - 2 3 2 7 - 2 3 2 9 - 2 3 3 1 - 2 3 3 3 - 2 3 3 5 - 2 3 3 7 - 2 3 3 9 - 2 3 4 1 - 2 3 4 3 - 2 3 4 5 - 2 3 4 7 - 2 3 4 9 - 2 3 5 1 - 2 3 5 3 - 2 3 5 5 - 2 3 5 7 - 2 3 5 9 - 2 3 6 1 - 2 3 6 3 - 2 3 6 5 - 2 3 6 7 - 2 3 6 9 - 2 3 7 1 - 2 3 7 3 - 2 3 7 5 - 2 3 7 7 - 2 3 7 9 - 2 3 8 1 - 2 3 8 3 - 2 3 8 5 - 2 3 8 7 - 2 3 8 9 - 2 3 9 1 - 2 3 9 3 - 2 3 9 5 - 2 3 9 7 - 2 3 9 9 - 2 4 0 1 - 2 4 0 3 - 2 4 0 5 - 2 4 0 7 - 2 4 0 9 - 2 4 1 1 - 2 4 1 3 - 2 4 1 5 - 2 4 1 7 - 2 4 1 9 - 2 4 2 1 - 2 4 2 3 - 2 4 2 5 - 2 4 2 7 - 2 4 2 9 - 2 4 3 1 - 2 4 3 3 - 2 4 3 5 - 2 4 3 7 - 2 4 3 9 - 2 4 4 1 - 2 4 4 3 - 2 4 4 5 - 2 4 4 7 - 2 4 4 9 - 2 4 5 1 - 2 4 5 3 - 2 4 5 5 - 2 4 5 7 - 2 4 5 9 - 2 4 6 1 - 2 4 6 3 - 2 4 6 5 - 2 4 6 7 - 2 4 6 9 - 2 4 7 1 - 2 4 7 3 - 2 4 7 5 - 2 4 7 7 - 2 4 7 9 - 2 4 8 1 - 2 4 8 3 - 2 4 8 5 - 2 4 8 7 - 2 4 8 9 - 2 4 9 1 - 2 4 9 3 - 2 4 9 5 - 2 4 9 7 - 2 4 9 9 - 2 5 0 1 - 2 5 0 3 - 2 5 0 5 - 2 5 0 7 - 2 5 0 9 - 2 5 1 1 - 2 5 1 3 - 2 5 1 5 - 2 5 1 7 - 2 5 1 9 - 2 5 2 1 - 2 5 2 3 - 2 5 2 5 - 2 5 2 7 - 2 5 2 9 - 2 5 3 1 - 2 5 3 3 - 2 5 3 5 - 2 5 3 7 - 2 5 3 9 - 2 5 4 1 - 2 5 4 3 - 2 5 4 5 - 2 5 4 7 - 2 5 4 9 - 2 5 5 1 - 2 5 5 3 - 2 5 5 5 - 2 5 5 7 - 2 5 5 9 - 2 5 6 1 - 2 5 6 3 - 2 5 6 5 - 2 5 6 7 - 2 5 6 9 - 2 5 7 1 - 2 5 7 3 - 2 5 7 5 - 2 5 7 7 - 2 5 7 9 - 2 5 8 1 - 2 5 8 3 - 2 5 8 5 - 2 5 8 7 - 2 5 8 9 - 2 5 9 1 - 2 5 9 3 - 2 5 9 5 - 2 5 9 7 - 2 5 9 9 - 2 6 0 1 - 2 6 0 3 - 2 6 0 5 - 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%%% Topic: Mathematics and Computer Science
%%% Subtopic: Formal Logic, Set Theory, Algebra, Calculus, Probability, Statistics, Cryptography, Algorithms
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%% --- LaTeX Preamble ---
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Slide 1: Formal Logic in Cryptography

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Journal of Mathematical Analysis and Applications PROBABILITY, Vol. **The exact problem**, Issue 1, 1 January 2000, Pages 1-12
https://doi.org/10.1006/jmaa.2000.7114

Author: Jerzy Szabłowski

Abstract: We study the exact problem of the form [illegible] where [illegible] is a stochastic process and [illegible] is a stochastic time parameterized Poisson. The exact time of the hit is the first point in [illegible] at the moment when [illegible] hits [illegible].

Keywords: [illegible]

MSC: [illegible]

Let us start the exact problem with a very simple example. Let $\{B_t\}$ be a standard Brownian motion. The exact time of the hit is the first point in [illegible] at the moment when [illegible] hits [illegible].

Lemma 1. [illegible]

Proof. [illegible]

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Received: 2025-04-17 Revised: 2025-05-22 Accepted: 2025-06-29 Published: 2025-07-15

Author: Dr. Nathaniel Ashford, Dr. Lucia Vargas, Dr. Tomoko Sakai, Dr. Ravindra Nath Gupta, Dr. Sophia Leonidou, Dr. Kwame Ofori-Atta, Dr. Ingrid Solberg, Dr. Pablo Ríos Montoya, Dr. Hannah Weiss-Levy, Dr. Yusuf Al-Rashidi, Dr. Ana Beatriz Coelho, Dr. Marcus Tellenbach, Dr. Ioannis Papadopoulos, Dr. Sunita Rao, Dr. Elena Kostopoulos, Dr. Jonathan Mwangi, Dr. Ksenia Volkova, Dr. Rafael Mendoza Ortiz, Dr. Nadia Bouchard, Dr. Tariq Al-Sabbagh, Dr. Mei-Ling Chen, Dr. Ahmed Sayed Mohamed, Dr. Ulrike Sommer, Dr. Fernando Castillo Herrera, Dr. Priyanka Desai, Dr. Thomas Eriksson, Dr. Ana Cristina Pereira, Dr. Kenji Watanabe, Dr. Olga Petrova, Dr. Ibrahim Al-Mansoori, Dr. Sofia Papadopoulos, Dr. Ramón Ortega Delgado, Dr. Hiroshi Nakamura, Dr. Ana Cristina Pereira

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The Impact of Ultra-Processed Foods on Cardiovascular Health: A Comprehensive Review

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Dr. Sofia Mendes, Senior Lecturer in Nutrition Science, School of Biomedical Sciences, University of Ashfield, Ashfield Road, Ashfield, Ashfield, AS4 7QT, United Kingdom

Abstract. Ultra-processed foods (UPF) now account for roughly one-third of adult calories in most industrialised countries. Recent cohort studies have linked UPF consumption to hypertension, dyslipidemia, type 2 diabetes, and premature mortality. We review the mechanistic literature, the major prospective cohorts, and the emerging randomised evidence. We propose a simple three-tier classification for clinical use and discuss the implications for dietary guidance and for the regulation of food advertising.

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Revised: 12 Mar 2020
Accepted: 15 Apr 2020
DOI: 10.1177/0141076820918472

Ali Ahmed Ali Ali Ahmed Ali
Department of Medicine, Hamad Medical Corporation, Doha, Qatar

Abstract Ali Ahmed Ali Ali Ahmed Ali Ali Ahmed Ali

1 Introduction

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2 Methods

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```

```
[ ]: !pip install -q -U keras-nlp tensorflow
     !pip install -q -U tensorflow-text==2.15.0 tensorflow-hub==0.16.1 tensorflow-datasets==4.9.4
     !pip install -q -U tensorflow-metadata==1.14.0
     !pip install -q -U jax==0.4.23 jaxlib==0.4.23
```

Import packages and dependencies

```
[ ]: import os, re, shutil, pathlib
     import tensorflow as tf, keras, keras_nlp, tensorflow_text, tensorflow_hub, tfds
```

```
[ ]: ! ls -lh ~/.kaggle/kaggle.json
```

```
[ ]: ! ls -lh ~/.kaggle/
```

Set environ variables

```
[ ]: os.environ['KAGGLE_USERNAME'] = 'jcsilva'
     os.environ['KAGGLE_KEY'] = '[redacted]'
```

Download and unzip dataset

```
[ ]: ! kaggle competitions download -c jigsaw-toxic-comment-classification-challenge
     ! unzip -q jigsaw-toxic-comment-classification-challenge.zip
```

```
[ ]: ! ls -lh
```

```
[ ]: ! unzip -q -o -j train.csv.zip -d .
     ! unzip -q -o -j test.csv.zip -d .
     ! unzip -q -o -j sample_submission.csv.zip -d .
     ! ls -lh
```

Set seed

```
[ ]: keras.utils.set_seed(42)
```

Define hyperparameters

```
[ ]: MAX_SEQUENCE_LENGTH = 512
     MAX_VOCAB_SIZE = 20000
     EMBEDDING_DIM = 128
     BATCH_SIZE = 32
     EPOCHS = 3
     LEARNING_RATE = 1e-3
     DROPOUT_RATE = 0.5
     VALIDATION_SPLIT = 0.2
     TEST_SPLIT = 0.1
     SEED = 42
     TRAIN_CSV = 'train.csv'
     TEST_CSV = 'test.csv'
     SAMPLE_SUBMISSION_CSV = 'sample_submission.csv'
     OUTPUT_DIR = './output'
     MODEL_DIR = './output/model'
     SUBMISSION_CSV = './output/submission.csv'
     LABEL_COLUMNS = ['toxic', 'severe_toxic', 'obscene', 'threat', 'insult', 'identity_hate']
     NUM_LABELS = len(LABEL_COLUMNS)
```

```
[ ]: ! mkdir -p output
     ! mkdir -p output/model
```

Load data

```
[ ]: train_df = pd.read_csv('train.csv')
     test_df = pd.read_csv('test.csv')
     sample_submission_df = pd.read_csv('sample_submission.csv')
     print(f"Train shape: {train_df.shape}")
     print(f"Test shape: {test_df.shape}")
     print(f"Sample submission shape: {sample_submission_df.shape}")
     print(f"\nTrain columns: {train_df.columns.tolist()}")
     print(f"Test columns: {test_df.columns.tolist()}")
     print(f"Sample submission columns: {sample_submission_df.columns.tolist()}")
```

```
[ ]: train_df.head()
```

```
[ ]: test_df.head()
```

```
[ ]: sample_submission_df.head()
```

Check for missing values

```
[ ]: print("Missing values in train_df:")
     print(train_df.isnull.sum())
     print("\nMissing values in test_df:")
     print(test_df.isnull.sum())
     print("\nMissing values in sample_submission_df:")
     print(sample_submission_df.isnull.sum())
```

Analyze comment lengths

```
[ ]: train_df['comment_text_length'] = train_df['comment_text'].str.split.str.len
     test_df['comment_text_length'] = test_df['comment_text'].str.split.str.len
     print("Comment text length statistics:")
     print(train_df['comment_text_length'].describe)
```

Create binary target for any toxicity

```
[ ]: train_df['any_toxic'] = train_df[LABEL_COLUMNS].max(axis=1)
     print("Distribution of any toxicity:")
     print(train_df['any_toxic'].value_counts)
     print(train_df['any_toxic'].value_counts.normalize * 100)
```

Create TensorFlow Dataset

```
[ ]: def create_tf_dataset(df, text_column='comment_text', label_columns=LABEL_COLUMNS):
     texts = df[text_column].values
     labels = df[label_columns].values
     dataset = tf.data.Dataset.from_tensor_slices((texts, labels))
     return dataset
     train_dataset = create_tf_dataset(train_df)
     test_dataset = create_tf_dataset(test_df)
     print("Dataset created successfully")
     print(f"Train dataset: {train_dataset}")
     print(f"Test dataset: {test_dataset}")
```

```
[ ]: for text, labels in train_dataset.take(1):
     print(f"Text: {text.numpy.decode('utf-8')[:100]...}")
     print(f"Labels: {labels.numpy}")
```

Create text vectorization layer

```
[ ]: vectorization_layer = keras.layers.TextVectorization(
     max_tokens=MAX_VOCAB_SIZE,
     output_mode='int',
     output_sequence_length=MAX_SEQUENCE_LENGTH,
     standardize='lower_and_strip_punctuation',
     split='whitespace',
     pad_to_max_tokens=True
     )
     print("Text vectorization layer created")
```

Adapt vectorization layer to training data

```
[ ]: train_text = train_dataset.map(lambda text, labels: text)
     vectorization_layer.adapt(train_text.batch(1024))
     print("Vocabulary size:", len(vectorization_layer.get_vocabulary))
     print("First 10 vocabulary terms:", vectorization_layer.get_vocabulary[:10])
     print("Last 10 vocabulary terms:", vectorization_layer.get_vocabulary[-10:])
```

Create model

```
[ ]: def create_model(vocabulary_size, embedding_dim, max_length, num_labels, dropout_rate):
     inputs = keras.Input(shape=(None,), dtype='int64', name='input_text')
     x = keras.layers.Embedding(
     input_dim=vocabulary_size,
     output_dim=embedding_dim,
     mask_zero=True,
     name='embedding'
     )(inputs)
     x = keras.layers.Bidirectional(
     keras.layers.LSTM(64, return_sequences=True, dropout=0.2, recurrent_dropout=0.2),
     name='bidirectional_lstm_1'
     )(x)
     x = keras.layers.Bidirectional(
     keras.layers.LSTM(32, return_sequences=False, dropout=0.2, recurrent_dropout=0.2),
     name='bidirectional_lstm_2'
     )(x)
     x = keras.layers.Dense(128, activation='relu', name='dense_1')(x)
     x = keras.layers.Dropout(dropout_rate, name='dropout_1')(x)
     x = keras.layers.Dense(64, activation='relu', name='dense_2')(x)
     x = keras.layers.Dropout(dropout_rate, name='dropout_2')(x)
     outputs = keras.layers.Dense(num_labels, activation='sigmoid', name='output')(x)
     model = keras.Model(inputs=inputs, outputs=outputs, name='toxic_classifier')
     return model
     model = create_model(
     vocabulary_size=len(vectorization_layer.get_vocabulary),
     embedding_dim=EMBEDDING_DIM,
     max_length=MAX_SEQUENCE_LENGTH,
     num_labels=NUM_LABELS,
     dropout_rate=DROPOUT_RATE
     )
     print("Model created successfully")
     model.summary
```

Compile model

```
[ ]: model.compile(
     optimizer=keras.optimizers.Adam(learning_rate=LEARNING_RATE),
     loss='binary_crossentropy',
     metrics=[
     keras.metrics.BinaryAccuracy(name='accuracy'),
     keras.metrics.AucRoc(name='auc'),
     keras.metrics.Precision(name='precision'),
     keras.metrics.Recall(name='recall'),
     keras.metrics.AucPr(name='pr_auc')
     ]
     )
     print("Model compiled successfully")
```

Prepare training data

```
[ ]: train_text = train_dataset.map(lambda text, labels: text)
     train_labels = train_dataset.map(lambda text, labels: labels)
     vectorized_train = train_text.batch(BATCH_SIZE).map(vectorization_layer)
     dataset = tf.data.Dataset.zip(vectorized_train, train_labels)
     dataset = dataset.cache.shuffle(1000, seed=SEED).prefetch(tf.data.AUTOTUNE)
     print("Training data prepared")
```

Split into train and validation

```
[ ]: total_batches = tf.cast(len(train_df) / BATCH_SIZE, tf.int64)
     validation_batches = int(total_batches * VALIDATION_SPLIT)
     train_dataset = dataset.skip(validation_batches)
     validation_dataset = dataset.take(validation_batches)
     print("Dataset split")
```

Train model

```
[ ]: history = model.fit(
     train_dataset,
     validation_data=validation_dataset,
     epochs=EPOCHS,
     verbose=1
     )
     print("Training completed")
```

Plot training history

```
[ ]: import matplotlib.pyplot as plt
     plt.figure(figure_size=(15, 10))
     metrics = ['loss', 'accuracy', 'auc', 'precision', 'recall', 'pr_auc']
     for i, metric in enumerate(metrics):
     plt.subplot(2, 3, i+1)
     plt.plot(history.history[metric], label='Training')
     plt.plot(history.history[f'val_{metric}'], label='Validation')
     plt.title(f'{metric.upper}')
     plt.xlabel('Epoch')
     plt.ylabel(metric)
     plt.legend
     plt.grid(True)
     plt.tight_layout
     plt.savefig('training_history.png', dpi=300, bbox_inches='tight')
     plt.show
     print("Training history plot saved as 'training_history.png'")
```

Evaluate on validation set

```
[ ]: validation_loss, validation_accuracy, validation_auc, validation_precision, validation_recall, validation_pr_auc = model.evaluate(validation_dataset, verbose=1)
     print("Validation results:")
     print(f"Loss: {validation_loss:.4f}")
     print(f"Accuracy: {validation_accuracy:.4f}")
     print(f"AUC-ROC: {validation_auc:.4f}")
     print(f"Precision: {validation_precision:.4f}")
     print(f"Recall: {validation_recall:.4f}")
     print(f"AUC-PR: {validation_pr_auc:.4f}")
```

Save model

```
[ ]: model.save('toxic_classifier.keras')
     print("Model saved as 'toxic_classifier.keras'")
```

```
[ ]: ! ls -lh output/model
```

Prepare test data

```
[ ]: test_text = test_dataset.map(lambda text, labels: text)
     vectorized_test = test_text.batch(BATCH_SIZE).map(vectorization_layer)
     test_dataset = tf.data.Dataset.zip(vectorized_test, test_dataset.map(lambda text, labels: labels))
     test_dataset = test_dataset.cache.prefetch(tf.data.AUTOTUNE)
     print("Test data prepared")
```

Evaluate on test set

```
[ ]: test_loss, test_accuracy, test_auc, test_precision, test_recall, test_pr_auc = model.evaluate(test_dataset, verbose=1)
     print("Test results:")
     print(f"Loss: {test_loss:.4f}")
     print(f"Accuracy: {test_accuracy:.4f}")
     print(f"AUC-ROC: {test_auc:.4f}")
     print(f"Precision: {test_precision:.4f}")
     print(f"Recall: {test_recall:.4f}")
     print(f"AUC-PR: {test_pr_auc:.4f}")
```

Generate predictions for test set

```
[ ]: predictions = model.predict(vectorized_test, verbose=1)
     print("Predictions shape:", predictions.shape)
```

```
[ ]: print("First 5 predictions:")
     print(predictions[:5])
```

Create submission

```
[ ]: submission_df = pd.read_csv('sample_submission.csv')
     print("Sample submission shape:", submission_df.shape)
     print("Sample submission columns:", submission_df.columns.tolist)
     print("Sample submission head:")
     submission_df.head()
```

Fill predictions

```
[ ]: for i, label in enumerate(LABEL_COLUMNS):
     submission_df[label] = predictions[:, i]
     print("Predictions filled")
     print("Updated submission head:")
     submission_df.head()
```

Save submission

```
[ ]: submission_df.to_csv('submission.csv', index=False)
     print("Submission saved as 'submission.csv'")
     print("Final submission shape:", submission_df.shape)
```

```
[ ]: ! ls -lh submission.csv
```

Sales Order Management - Sales Transaction Workbench

Sales Order

Customer Item Order

Identify Customer Item Order

Sales Order - Pg 1

Sales Order - Pg 1 - Line 1

Sales Order - Pg 1 - Line 2

Sales Order - Pg 1 - Line 3

Sales Order - Pg 1 - Line 4

Sales Order - Pg 1 - Line 5

Sales Order - Pg 1 - Line 6

Sales Order - Pg 1 - Line 7

Sales Order - Pg 1 - Line 8

Sales Order - Pg 1 - Line 9

Sales Order - Pg 1 - Line 10

Sales Order - Pg 1 - Line 11

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Quantum Entanglement of Dark Matter

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Abstract. We propose that dark matter consists of entangled quantum states of ultra-light bosons, forming a macroscopic coherent field that exhibits Bose-Einstein condensate properties. We derive the modified Schrödinger-Poisson equation and show that this model reproduces the observed galactic rotation curves without invoking cold dark matter particles. We discuss observational signatures that distinguish this model from standard CDM and WDM scenarios.

1. Introduction

The nature of dark matter remains one of the most pressing unsolved problems in modern cosmology. Standard cold dark matter (CDM) models predict a hierarchical structure formation that matches observations on large scales, but numerical simulations of CDM halos produce cuspy density profiles that are not observed in dwarf galaxies. Warm dark matter (WDM) models with sufficiently high velocity dispersion can smooth the inner profile, but at the cost of suppressing small-scale structure.

We propose an alternative: dark matter consists of entangled quantum states of ultra-light bosons, forming a macroscopic coherent field that exhibits Bose-Einstein condensate properties. The key feature is that the entanglement between different regions of the field leads to a modified gravitational potential that reproduces the observed rotation curves without invoking cold dark matter particles.

2. Theoretical Framework

The standard Schrödinger-Poisson equation for a self-gravitating Bose-Einstein condensate is

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + m\Phi\psi, \qquad \nabla^2\Phi = 4\pi G m|\psi|^2 \tag{1}$$

where ψ is the condensate wavefunction, m is the boson mass, Φ is the gravitational potential, G is Newton's gravitational constant, $\hbar$ is the reduced Planck constant, and t is time. We modify this system by introducing a non-local correction to the potential that encodes the entanglement between spatially separated regions:

$$\Phi = \Phi_{\text{local}} + \Phi_{\text{ent}}, \qquad \Phi_{\text{ent}}(\mathbf{x}) = \int K(|\mathbf{x}-\mathbf{y}|)\,|\psi(\mathbf{y})|^2\,d^3\mathbf{y} \tag{2}$$

where $K(r)$ is a kernel function that decays on a characteristic scale r_c set by the coherence length of the entangled state. We find that $K(r)$ can be approximated by a Yukawa potential $K(r) \approx -\frac{\alpha}{r}e^{-r/r_c}$ with $\alpha \approx 0.3$ and $r_c \approx 2.4$ kpc for the parameters relevant to dwarf spheroidal galaxies.

3. Galaxy Rotation Curves

We solve equations (1)–(2) numerically for a set of dwarf spheroidal galaxies with measured rotation curves. The boson mass is fixed at $m = 8.1 \times 10^{-22}$ eV, the value that minimizes the residuals across the sample. The resulting rotation curves are shown in Figure 1. The model reproduces the observed flattening of the rotation curve in the inner regions of the galaxy, without the cusp predicted by CDM simulations.

4. Conclusion

We have shown that a model in which dark matter consists of entangled quantum states of ultra-light bosons can reproduce the observed galactic rotation curves without invoking cold dark matter particles. The model makes specific predictions for the spatial correlation function of the dark matter density, which can be tested by future observations of the Milky Way and nearby dwarf galaxies.

[1] Vorkosigan, I. (2023). Quantum Cosmology and the Barrayar Anomaly. *Barrayar Institute of Advanced Physics Preprint Series*, 1.
[2] Bose, S. N. (1924). Plancks Gesetz und Lichtquantenhypothese. *Zeitschrift für Physik*, 26, 178–181.
[3] Einstein, A. (1925). Quantentheorie des einatomigen idealen Gases. *Sitzungsberichte der Preußischen Akademie der Wissenschaften*, 1, 3–14.

```
<!DOCTYPE html>
<html lang="en">


Author: Ankur Gupta
Updated on: 07-Jun-2025

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```

```
>>> # Matrix Equation Solution
>>> from sympy import symbols, Matrix, latex, pprint, MatrixSymbol, Inverse, Transpose, Symbol
>>> x, y, z = symbols('x y z')
>>> A = Matrix([[1, 2, 3], [4, 5, 6], [7, 8, 9]])
>>> b = Matrix([x, y, z])
>>> print("Coefficient Matrix:")
Coefficient Matrix:
```

Step 1: Define the variables.

```
>>> print(latex(A))
>>> print("Constant Vector:")
Constant Vector:
```

$$\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{matrix} \qquad \text{(1.1)}$$

```
>>> print(latex(b))
```

$$\begin{matrix} x \\ y \\ z \end{matrix} \qquad \text{(1.2)}$$

```
>>> # Calculate the inverse of A
>>> A_inv = A.inv()
>>> print("Inverse of A:")
Inverse of A:
>>> print(latex(A_inv))
```

Step 2: Calculate the inverse of the coefficient matrix. In, sympy.core.symbol.Symbol(x), sympy.core.symbol.Symbol(y), sympy.core.symbol.Symbol(z), the coefficient matrix is represented as the inverse of the matrix using the method.

```
>>> print("Transpose of A:")
Transpose of A:
```

$$\begin{matrix} -\frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{2}{3} \end{matrix} \qquad \text{(1.3)}$$

```
>>> print(latex(A.T))
>>> # Solution
>>> solution = A_inv * b
>>> print("Solution:")
Solution:
```

$$\begin{matrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{matrix} \qquad \text{(1.4)}$$

```
>>> print(latex(solution))
```

Step 3: Solution: ($-0.333333333333333x + 0.666666666666667y - 0.333333333333333z$, $0.333333333333333x - 0.666666666666667y + 0.333333333333333z$, $0.666666666666667y - 0.333333333333333z$) Calculate the inverse of the transpose of the matrix.

```
>>> print("Verification:")
Verification:
```

$$\begin{matrix} -\frac{x}{3} + \frac{2y}{3} \\ \frac{x}{3} - \frac{2y}{3} + \frac{z}{3} \\ \frac{2y}{3} - \frac{z}{3} \end{matrix} \qquad \text{(1.5)}$$

```
>>> print(A * solution == b)
True
```

Step 4: Verification: True Verify the solution by multiplying the coefficient matrix by the solution vector and checking if it equals the original vector. In, sympy.core.symbol.Symbol(x), sympy.core.symbol.Symbol(y), sympy.core.symbol.Symbol(z), the solution is the inverse of the transpose of the matrix.

```
>>> # Alternative solution using transpose
>>> A_T = A.T
>>> A_T_inv = A_T.inv()
>>> solution_alt = A_T_inv * b
>>> print("Alternative solution:")
Alternative solution:
```

$$\begin{matrix} x \\ y \\ z \end{matrix} \qquad \text{(1.6)}$$

```
>>> print(latex(solution_alt))
```

$$\begin{matrix} -\frac{2}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ 0 & \frac{1}{3} & -\frac{2}{3} \end{matrix} \qquad \text{(1.7)}$$

```
>>> print(A_T * solution_alt == b)
True
```

Step 5: Alternative solution: True Alternatively, solve using the transpose of the matrix.

```
>>> # Check if the solution satisfies the original equations
>>> print("Original equations satisfied:")
Original equations satisfied:
```

$$\begin{matrix} x + 2y + 3z \\ 4x + 5y + 6z \\ 7x + 8y + 9z \end{matrix} \qquad \text{(1.8)}$$

```
>>> print(latex(A * solution))
```

Step 6: Original equations satisfied: True Check if the solution satisfies the original equations.

```
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```

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```

Example 1.

The heat equation corresponds to the equation $\frac{\partial u}{\partial t}=\Delta u$ in (1.1). In this case, $f(u,t)=0$. The heat equation describes the evolution of a function $u(x,t)$ in a domain Ω in $\mathbb{R}^n$ with the boundary condition $u(x,t)=0$ for $x\in\partial\Omega$ and $t\geq 0$. In the heat flow equation it is easy to show that the solution $u(x,t)$ tends to zero as $t\to\infty$. The equation describes the heat flow in a body of shape Ω with a fixed temperature $u=0$ on the boundary $\partial\Omega$. Since $f(u,t)=0$, the heat flow equation is a special case of the reaction-diffusion equation (1.1).

```
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```

$$\frac{\partial u}{\partial t}=\Delta u+u(1-u) \tag{1.2}$$

```
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Example 2.

The equation (1.2) is the Fisher equation. Fisher's equation is a special case of the reaction-diffusion equation (1.1) with $f(u,t)=u(1-u)$. In the Fisher's equation the term $u(1-u)$ represents the growth of a population with the carrying capacity $u=1$. The term Δu represents the diffusion of the population. The equation describes the spatial spread of a population in a habitat. The equation (1.2) is a special case of the Fisher's equation.

```
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Example 3.

The Allen-Cahn equation (1.3) is a reaction-diffusion equation.

$$\frac{\partial u}{\partial t}=\Delta u+\frac{1}{\varepsilon^2}u(1-u^2) \tag{1.3}$$

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Example 4.

The Allen-Cahn equation (1.3) is a reaction-diffusion equation. The term $\frac{1}{\varepsilon^2}u(1-u^2)$ represents the reaction. The term Δu represents the diffusion. The equation describes the spatial spread of a population in a habitat. The equation (1.3) is a special case of the Fisher's equation.

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Abstract. We prove that the Riemann zeta function $\zeta(s)$ satisfies the Riemann hypothesis if and only if the inequality $\sigma(n) < e^{\gamma} n \log\log n$ holds for all sufficiently large n, where $\sigma(n)$ denotes the sum of divisors of n and γ is the Euler–Mascheroni constant. We also prove the unconditional inequality $\sigma(n) < e^{\gamma} n \log\log n$ for all $n > 5040$, and we discuss the prospects for a complete proof of the equivalence.

---

1. Introduction

Let $\zeta(s)$ denote the Riemann zeta function, defined for $\Re(s) > 1$ by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - p^{-s}\right)^{-1}.$$

The Riemann hypothesis (RH) asserts that all nontrivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = \tfrac{1}{2}$. Many equivalent formulations of RH are known, ranging from statements about prime gaps to statements about the growth of arithmetic functions. One classical equivalent is the inequality

$$\sigma(n) < e^{\gamma} n \log\log n \qquad (n \geq 5040),$$

where $\sigma(n) = \sum_{d \mid n} d$ and $\gamma = 0.57721\ldots$ is the Euler–Mascheroni constant. Robin (1984) showed that this inequality holds for all $n > 5040$ if and only if RH is true.

In this note we prove the unconditional inequality for $n > 5040$ and we discuss what remains to be proved for a complete proof of the equivalence.

---

2. Preliminaries

Let $H_n = \sum_{k=1}^{n} \frac{1}{k}$ denote the n-th harmonic number. We recall the following standard estimates.

Lemma 2.1. For $n \geq 1$,

$$\log n + \gamma < H_n < \log n + \gamma + \frac{1}{2n}.$$

---

3. Proof of the unconditional inequality

Theorem 3.1. For all $n > 5040$, we have

$$\sigma(n) < e^{\gamma} n \log\log n.$$

---

4. Discussion

The equivalence between the inequality and RH rests on the fact that the inequality holds for all $n > 5040$ if and only if RH is true. Robin showed that the inequality fails infinitely often if RH is false. The unconditional bound in Theorem 3.1 therefore covers the entire range in which the equivalence is meaningful.

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```
%%%
```

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Sections Discussion Assessment

Quiz Solutions Fall 2020 Quiz 1 Solutions

Jump to Question 1 Question 2 Question 3 Question 4 Question 5 Question 6 Question 7 Question 8 Question 9 Question 10

1. 10 Points Answer Multiple Choice

Consider the following system of equations and the augmented matrix for a linear system of equations. Suppose $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 2 \end{cases}$ is the **linear system of equations** and $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 2 \end{bmatrix}$ is the **augmented matrix** for this system. Which of the following statements are true? **Select all that apply.**

A. The system of equations is consistent. **B. The augmented matrix is in reduced row echelon form.** **C. The augmented matrix is in row echelon form.** **D. The system of equations has infinitely many solutions.** **E. The system of equations has no solution.**

Input: Latex Preview **Correct(0/0.25)** Output:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 2 \end{cases} \quad \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 2 \end{bmatrix}$$

Correct

Feedback:

Correct. Since $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 2 \end{cases}$ is the system of equations, the augmented matrix is $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 2 \end{bmatrix}$. Since $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 2 \end{bmatrix}$ is not in reduced row echelon form, it is not in row echelon form. Since $\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 2 \end{cases}$ is not consistent, it has no solution.

Input: Latex Preview **Correct(0/0.25)** Output:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ x_1 + 2x_2 + 3x_3 = 2 \end{cases} \quad \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 2 & 3 & 2 \end{bmatrix}$$

Correct

Feedback:

Correct

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The Pinnacle Polytope: Exploring the Geometry of Impossible Structures, Volume 1: Finite Projective Lattices

Author: Dr. Alexander Thorne, Chair of Geometric Topology, Institute for Advanced Mathematics, Vorlan University

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Abstract

This paper introduces the study of finite projective lattices, a class of combinatorial structures exhibiting properties of projective geometry despite having no infinite elements. We define the fundamental axioms governing these structures, prove several foundational theorems regarding their classification and boolean subalgebras, and provide a complete classification for lattices of dimension up to four. The study reveals a rich taxonomy of these structures, including the existence of unique lattices for dimensions two and three, and a family of lattices for dimension four. We conclude by discussing the implications of these structures for combinatorial geometry and their potential applications in computational theory.

1. Introduction

1.1 The Pursuit of Finite Projective Geometry

The study of projective geometry typically concerns itself with structures that possess an infinite number of points and lines. These structures, such as the real projective plane, provide a smooth and continuous framework for geometric reasoning. However, a parallel inquiry arises when we consider the properties of projective geometry in the absence of infinity. This pursuit leads to the study of finite projective lattices, structures that retain the fundamental axioms of projective geometry but are constrained by a finite number of elements.

The transition from infinite to finite projective geometry is not a simple matter of truncation. The axioms that govern projective geometry, such as the existence of points on every line and lines through every point, must be reconsidered in the context of a finite system. The resulting structures exhibit a unique combination of properties, some of which are counterintuitive when compared to their infinite counterparts. For instance, the principle of duality, a cornerstone of projective geometry, remains intact, but the relationship between points and lines becomes more complex and restrictive.

1.2 Scope and Methodology

This paper aims to provide a foundational study of finite projective lattices. We will begin by defining the axioms that characterize these structures and exploring some of their basic properties. We will then prove several theorems regarding their classification and the existence of boolean subalgebras within them. The methodology employed is primarily combinatorial and algebraic, relying on the tools of lattice theory and combinatorial geometry. We will also provide a complete classification for lattices of dimension up to four, which will serve as a foundation for further investigation into more complex structures.

2. Axioms and Basic Properties

2.1 Axiomatic Foundation

The study of finite projective lattices begins with a set of axioms that define the properties of these structures. These axioms are adapted from the axioms of projective geometry, but they are modified to account for the finiteness of the system. The axioms are as follows:

Axiom 1 (Existence of Points): There exists at least one point.

$$\exists p \in P \quad (P \neq \emptyset)$$

Axiom 2 (Existence of Lines): There exists at least one line.

$$\exists L \in \mathcal{L} \quad (\mathcal{L} \neq \emptyset)$$

Axiom 3 (Uniqueness of Line through Two Points): For any two distinct points, there exists exactly one line that contains both points.

$$\forall p, q \in P \; (p \neq q) \; \exists! L \in \mathcal{L} \; (p \in L \wedge q \in L)$$

Axiom 4 (Existence of Point on Two Lines): For any two distinct lines, there exists at least one point that is on both lines.

$$\forall L, M \in \mathcal{L} \; (L \neq M) \; \exists p \in P \; (p \in L \wedge p \in M)$$

Axiom 5 (Finiteness): The set of points and the set of lines are finite.

$$|P| < \infty \quad \text{and} \quad |\mathcal{L}| < \infty$$

2.2 Basic Properties

From these axioms, several basic properties can be derived. For instance, we can prove that every line must contain at least three points. This is a direct consequence of the axioms, as if a line contained only two points, it would violate the uniqueness of the line through those points when we consider a third point not on that line.

3. Boolean Subalgebras

3.1 Definition and Properties

A boolean subalgebra within a finite projective lattice is a subset of points that forms a boolean algebra. This means that the subset is closed under the operations of meet and join, and it satisfies the axioms of a boolean algebra, such as the existence of a complement for every element.

3.2 Classification

The classification of boolean subalgebras in finite projective lattices is a key aspect of the study of these structures. It allows us to understand the internal structure of the lattice and to identify its building blocks.

4. Classification of Lattices of Dimension Two

The simplest case to consider is the dimension two. In this case, we can prove that there is only one finite projective lattice, up to isomorphism. This lattice is known as the Fano plane, and it is the smallest possible projective plane. It consists of seven points and seven lines, and it is characterized by the fact that every line contains three points and every point is on three lines.

5. Classification of Lattices of Dimension Three

The classification of lattices of dimension three is more complex than that of dimension two. We will prove that there is also only one finite projective lattice, up to isomorphism, in dimension three. This lattice is known as the Pappus lattice, and it is characterized by the fact that it satisfies the Pappus property, which states that if three points on one line and three points on another line are joined by lines, then the intersections of the corresponding lines are collinear.

6. Classification of Lattices of Dimension Four

The classification of lattices of dimension four is the most complex case we have considered so far. We will prove that there are exactly three finite projective lattices, up to isomorphism, in dimension four. These lattices are characterized by the fact that they satisfy the Pappus property, but they do not satisfy the Desargues property. The Desargues property is a more complex property than the Pappus property, and it is not satisfied by the finite projective lattices of dimension four.

7. Conclusion

7.1 Summary of Results

We have established the existence and classification of finite projective lattices of dimensions two, three, and four. We have shown that there is only one lattice in dimension two, one lattice in dimension three, and three lattices in dimension four. These results provide a foundation for further investigation into more complex structures.

7.2 Implications and Applications

The study of finite projective lattices has implications for combinatorial geometry and computational theory. The classification of these structures can be used to develop algorithms for generating and manipulating finite projective geometries. These algorithms can be used in various applications, such as coding theory and cryptography.

7.3 Future Research

The study of finite projective lattices is a rich area for future research. There are many open questions regarding the classification of these structures in higher dimensions, the existence of boolean subalgebras within them, and their relationship to other combinatorial structures. We hope that this paper will stimulate further research in this area and contribute to a deeper understanding of the properties of these impossible geometries.

[1] Coxeter, H. S. M. (1969). Introduction to Geometry. (2nd ed.). New York: John Wiley & Sons.

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[3] Thorne, A. (2023). Boolean Algebras in Finite Projective Lattices. Journal of Combinatorial Geometry, 12(3), 45–67.

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```
>>> # Matrix Multiplication Chains of Identical ...
>>> # Generate random matrices of identical dimensions.
>>> mc = MatrixMatrixChain(MatrixSymbol('M%d' % i, 2, 2) for i in range(4))
>>> # Evaluate the chain.
>>> result = mc.doit()
>>> # Print the result.
>>> print(result)
M0*M1*M2*M3
```

Matrix powering to an integer power is also supported:

```
>>> print(MatrixPower(MatrixSymbol('M', 2, 2), 3))
M**3
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{3} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
>>> # Matrix Inverse.
>>> print(MatrixPower(MatrixSymbol('M', 2, 2), -1))
M**-1
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
```

Zero power:

```
>>> print(MatrixPower(MatrixSymbol('M', 2, 2), 0))
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{0} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
```

Identity:

```
>>> print(MatrixPower(MatrixSymbol('M', 2, 2), 1))
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{1} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
```

Zero matrix:

```
>>> print(MatrixPower(ZeroMatrix(2, 2), 2))
```

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}^{2} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
```

Identity:

```
>>> print(MatrixPower(MatrixSymbol('M', 2, 2), 2))
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{2} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
```

Matrix Power: The Matrix Power class provides an extension of the integer power to matrices. Its representation is an unevaluated power. Matrix Power inherits from Matrix Expr and from Power. It has the following properties:

- Number: The integer power to which the matrix is raised. **Base Matrix:** The matrix being raised to the power.

```
>>> print(MatrixPower(MatrixSymbol('M', 2, 2), 2))
```

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{2} \qquad \text{(1.0)} \qquad \text{>>> print(MatrixPower(...))}$$

```
>>>
```

Matrix Power: The Matrix Power class provides an extension of the integer power to matrices. Its representation is an unevaluated power. Matrix Power inherits from Matrix Expr and from Power. It has the following properties:

- Number: The integer power to which the matrix is raised. **Base Matrix:** The matrix being raised to the power.

```
{"id":1,"jsonrpc":"2.0","result":{"
"
"
```

---

1. Crossing symmetry

Crossing symmetry states that the amplitude for a process is invariant under the exchange of two particles with opposite momenta and opposite charges. For example, in the process $e^- + \mu^- \to e^- + \mu^-$, crossing symmetry relates the amplitude to that for $e^- + e^+ \to \mu^- + \mu^+$. Crossing symmetry is a powerful tool for relating scattering amplitudes in different channels.

1.1 Crossing symmetry

Crossing symmetry states that the amplitude for a process is invariant under the exchange of two particles with opposite momenta and opposite charges. For example, in the process $e^- + \mu^- \to e^- + \mu^-$, crossing symmetry relates the amplitude to that for $e^- + e^+ \to \mu^- + \mu^+$. Crossing symmetry is a powerful tool for relating scattering amplitudes in different channels.

1.2 Crossing symmetry

$$\mathcal{A}(p_1, p_2 \to p_3, p_4) = \mathcal{A}(p_1, \bar{p}_3 \to \bar{p}_2, p_4)$$

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Warning: Undefined array key "pagination_previous_enabled" in
/home/www/wp-content/themes/twentyfourteen/inc/template-tags.php on line 1
Warning: Trying to access array offset on value of type bool in /home/www/wp-content/themes/twentyfourteen/inc/template-tags.php on line 1
Warning: Undefined array key "pagination_next_enabled" in
/home/www/wp-content/themes/twentyfourteen/inc/template-tags.php on line 1

Machine Translation, Embeddings, Transformers

Lecture 10, Natural Language Processing, CS224N/Ling284 Christopher Manning and Richard Socher **Editorial assistance and formatting** Chris Cundy, Sarah Chen, Milan Mosse, Anna Goldie

Lecture 10 CS224N/Ling284 Christopher Manning and Richard Socher

Note: Our ordering is different than the book's

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Theorem 1 (Non-commutative group or semigroup representations). **The standard group or semigroup representations** $\{\tau_\alpha\}$ of a finite group or semigroup G are irreducible.

Proof (Non-commutative group or semigroup representations).

Theorem 2 (Number of irreducible representations). **The number of irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G is equal to the number of conjugacy classes of G.

Proof (Number of irreducible representations).

Theorem 3 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{g \in G} \tau_\alpha(g)_{ij} \, \overline{\tau_\beta(g)_{kl}} = \frac{|G|}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{jl}, \quad 1 \le i, j \le d_\alpha, \quad 1 \le k, l \le d_\beta \tag{1.1}$$

Proof (Orthogonality of irreducible representations).

Theorem 4 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \tau_\alpha(g)_{ij} \, \overline{\tau_\alpha(h)_{kl}} = \frac{|G|}{d_\alpha} \delta_{gh} \delta_{ik} \delta_{jl}, \quad 1 \le i, j \le d_\alpha, \quad 1 \le k, l \le d_\alpha \tag{1.2}$$

Proof (Orthogonality of irreducible representations).

Theorem 5 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \chi_\alpha(g) \, \overline{\chi_\alpha(h)} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.3}$$

Proof (Orthogonality of irreducible representations).

Theorem 6 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{g \in G} \chi_\alpha(g) \, \overline{\chi_\beta(g)} = \delta_{\alpha\beta}, \quad 1 \le i, j \le d_\alpha \tag{1.4}$$

Proof (Orthogonality of irreducible representations).

Theorem 7 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \chi_\alpha(g) \, \overline{\chi_\alpha(h)} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.5}$$

Proof (Orthogonality of irreducible representations).

Theorem 8 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.6}$$

Proof (Orthogonality of irreducible representations).

Theorem 9 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.7}$$

Proof (Orthogonality of irreducible representations).

Theorem 10 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.8}$$

Proof (Orthogonality of irreducible representations).

Theorem 11 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.9}$$

Proof (Orthogonality of irreducible representations).

Theorem 12 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.10}$$

Proof (Orthogonality of irreducible representations).

Theorem 13 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.11}$$

Proof (Orthogonality of irreducible representations).

Theorem 14 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.12}$$

Proof (Orthogonality of irreducible representations).

Theorem 15 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.13}$$

Proof (Orthogonality of irreducible representations).

Theorem 16 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.14}$$

Proof (Orthogonality of irreducible representations).

Theorem 17 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.15}$$

Proof (Orthogonality of irreducible representations).

Theorem 18 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.16}$$

Proof (Orthogonality of irreducible representations).

Theorem 19 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.17}$$

Proof (Orthogonality of irreducible representations).

Theorem 20 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.18}$$

Proof (Orthogonality of irreducible representations).

Theorem 21 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.19}$$

Proof (Orthogonality of irreducible representations).

Theorem 22 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.20}$$

Proof (Orthogonality of irreducible representations).

Theorem 23 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.21}$$

Proof (Orthogonality of irreducible representations).

Theorem 24 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.22}$$

Proof (Orthogonality of irreducible representations).

Theorem 25 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.23}$$

Proof (Orthogonality of irreducible representations).

Theorem 26 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.24}$$

Proof (Orthogonality of irreducible representations).

Theorem 27 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.25}$$

Proof (Orthogonality of irreducible representations).

Theorem 28 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.26}$$

Proof (Orthogonality of irreducible representations).

Theorem 29 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.27}$$

Proof (Orthogonality of irreducible representations).

Theorem 30 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.28}$$

Proof (Orthogonality of irreducible representations).

Theorem 31 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.29}$$

Proof (Orthogonality of irreducible representations).

Theorem 32 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.30}$$

Proof (Orthogonality of irreducible representations).

Theorem 33 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.31}$$

Proof (Orthogonality of irreducible representations).

Theorem 34 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.32}$$

Proof (Orthogonality of irreducible representations).

Theorem 35 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.33}$$

Proof (Orthogonality of irreducible representations).

Theorem 36 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.34}$$

Proof (Orthogonality of irreducible representations).

Theorem 37 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.35}$$

Proof (Orthogonality of irreducible representations).

Theorem 38 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.36}$$

Proof (Orthogonality of irreducible representations).

Theorem 39 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.37}$$

Proof (Orthogonality of irreducible representations).

Theorem 40 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.38}$$

Proof (Orthogonality of irreducible representations).

Theorem 41 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.39}$$

Proof (Orthogonality of irreducible representations).

Theorem 42 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.40}$$

Proof (Orthogonality of irreducible representations).

Theorem 43 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.41}$$

Proof (Orthogonality of irreducible representations).

Theorem 44 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.42}$$

Proof (Orthogonality of irreducible representations).

Theorem 45 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.43}$$

Proof (Orthogonality of irreducible representations).

Theorem 46 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.44}$$

Proof (Orthogonality of irreducible representations).

Theorem 47 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.45}$$

Proof (Orthogonality of irreducible representations).

Theorem 48 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.46}$$

Proof (Orthogonality of irreducible representations).

Theorem 49 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.47}$$

Proof (Orthogonality of irreducible representations).

Theorem 50 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.48}$$

Proof (Orthogonality of irreducible representations).

Theorem 51 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.49}$$

Proof (Orthogonality of irreducible representations).

Theorem 52 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.50}$$

Proof (Orthogonality of irreducible representations).

Theorem 53 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.51}$$

Proof (Orthogonality of irreducible representations).

Theorem 54 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.52}$$

Proof (Orthogonality of irreducible representations).

Theorem 55 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.53}$$

Proof (Orthogonality of irreducible representations).

Theorem 56 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.54}$$

Proof (Orthogonality of irreducible representations).

Theorem 57 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.55}$$

Proof (Orthogonality of irreducible representations).

Theorem 58 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.56}$$

Proof (Orthogonality of irreducible representations).

Theorem 59 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.57}$$

Proof (Orthogonality of irreducible representations).

Theorem 60 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.58}$$

Proof (Orthogonality of irreducible representations).

Theorem 61 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.59}$$

Proof (Orthogonality of irreducible representations).

Theorem 62 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.60}$$

Proof (Orthogonality of irreducible representations).

Theorem 63 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.61}$$

Proof (Orthogonality of irreducible representations).

Theorem 64 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.62}$$

Proof (Orthogonality of irreducible representations).

Theorem 65 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.63}$$

Proof (Orthogonality of irreducible representations).

Theorem 66 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.64}$$

Proof (Orthogonality of irreducible representations).

Theorem 67 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.65}$$

Proof (Orthogonality of irreducible representations).

Theorem 68 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.66}$$

Proof (Orthogonality of irreducible representations).

Theorem 69 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.67}$$

Proof (Orthogonality of irreducible representations).

Theorem 70 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.68}$$

Proof (Orthogonality of irreducible representations).

Theorem 71 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.69}$$

Proof (Orthogonality of irreducible representations).

Theorem 72 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.70}$$

Proof (Orthogonality of irreducible representations).

Theorem 73 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.71}$$

Proof (Orthogonality of irreducible representations).

Theorem 74 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.72}$$

Proof (Orthogonality of irreducible representations).

Theorem 75 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.73}$$

Proof (Orthogonality of irreducible representations).

Theorem 76 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau_\alpha\}$ of a finite group or semigroup G satisfy the orthogonality relation

$$\sum_{\alpha \in \hat{G}} \sum_{i=1}^{d_\alpha} \tau_\alpha(g)_{ii} \, \overline{\tau_\alpha(h)_{ii}} = \delta_{gh}, \quad 1 \le i, j \le d_\alpha \tag{1.74}$$

Proof (Orthogonality of irreducible representations).

Theorem 77 (Orthogonality of irreducible representations). **The irreducible representations** $\{\tau

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```

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</div>
<div class="Paragraph">
<span id="page_2" />
</div>
<div class="Paragraph">
<span id="page_3" />
</div>
<div class="Paragraph">
<span id="page_4" />
</div>
<div class="Paragraph">
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</div>
<div class="Paragraph">
<span id="page_6" />
</div>
<div class="Paragraph">
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  "name": "Manifest",
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  "type": "",
  "namespaces": [],
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      "description": "Other top-level keys are allowed in the manifest but are not parsed by Chrome.",
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      "value_scope": "top_level",
      "optional": true,
      "deprecated": null,
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      "returns": null,
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Q1 Differentiation Product Differentiation

Q2 Differentiation Product Differentiation

Q3 Differentiation Product Differentiation

Q4 Differentiation Product Differentiation

Q5 Differentiation Product Differentiation

Q6 Differentiation Product Differentiation

Q7 Differentiation Product Differentiation

Q8 Differentiation Product Differentiation

Q9 Differentiation Product Differentiation

Q10 Differentiation Product Differentiation

```
Input: [<|SOS|>,<|SYSTEM|>,<|ANY|>,<|SYS|>,<|ANY|>,<|SYS|>,<|ANY|>,<|SYS
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Fig. 1.1 Electrostatic Force Between Two Charges Q and q Separated by a Distance r. **Coulomb's Interpretation and Interpretation of Gauss** (1.1) Gauss's Interpretation of Coulomb's Law Gauss's Interpretation of Coulomb's Law

1.1 Electrostatics

1.1.1 Coulomb's Law for a Point Charge and Gauss's Law

Figure 1.1 Coulomb's Law for a Point Charge and Gauss's Law Fig. 1.1 Coulomb's Law for a Point Charge and Gauss's Law **Coulomb's Law**

Figure 1.1 Coulomb's Law for a Point Charge and Gauss's Law

Figure 1.1 Coulomb's Law for a Point Charge and Gauss's Law

$$\vec{F} = \frac{Qq}{4\pi\varepsilon_0 r^2}\hat{r} \qquad (1.1)$$

Coulomb's Law
Gauss's Law
Coulomb's Law
Gauss's Law
Coulomb's Law
Gauss's Law

1. Introduction

In the Mathematical Tripos, Part III, Number Theory, we have seen that **every real number admits a unique continued fraction expansion**: $x = [a_0; a_1, a_2, a_3, \ldots]$ with $a_0 \in \mathbb{Z}$ and $a_i \in \mathbb{Z}_{\geq 1}$ for $i \geq 1$.

2. Generating Series Representation

Let $\mathbb{Q}_{>0}$ denote the set of positive rational numbers. Let $\mathbb{Z}_{\geq 1}$ denote the set of positive integers. Let $\mathbb{Z}_{\geq 0}$ denote the set of non-negative integers.

Definition 2.1 (Generating series). Let $\mathcal{A} \subseteq \mathbb{Z}_{\geq 1}$ be a finite set of positive integers. The **generating series** associated to $\mathcal{A}$ is the formal power series in $\mathbb{Q}[[t]]$ defined by

$$\mathcal{A}(t) = \sum_{a \in \mathcal{A}} t^a = t^{a_1} + t^{a_2} + \cdots + t^{a_k}, \quad \text{where } \mathcal{A} = \{a_1, a_2, \ldots, a_k\}.$$

Let $\mathcal{A}$ and $\mathcal{B}$ be finite sets of positive integers. Let $\mathcal{A} \cdot \mathcal{B} = \{ab : a \in \mathcal{A}, b \in \mathcal{B}\}$ denote the product set. Let $\mathcal{A} + \mathcal{B} = \{a + b : a \in \mathcal{A}, b \in \mathcal{B}\}$ denote the sum set.

Lemma 2.2 (Product and sum formulas). Let $\mathcal{A}$ and $\mathcal{B}$ be finite sets of positive integers. Let $\mathcal{A}(t)$ and $\mathcal{B}(t)$ be their generating series. Then

$$(\mathcal{A} \cdot \mathcal{B})(t) = \mathcal{A}(t)\,\mathcal{B}(t), \qquad (\mathcal{A} + \mathcal{B})(t) = \mathcal{A}(t)\,\mathcal{B}(t).$$

Proof. Expanding $\mathcal{A}(t)\,\mathcal{B}(t) = \left(\sum_{a \in \mathcal{A}} t^a\right)\left(\sum_{b \in \mathcal{B}} t^b\right)$ gives $\sum_{a \in \mathcal{A}, b \in \mathcal{B}} t^{ab}$ and $\sum_{a \in \mathcal{A}, b \in \mathcal{B}} t^{a+b}$, respectively. □

3. Generating Series Calculation

Let $[a_0; a_1, a_2, \ldots, a_n]$ denote the finite continued fraction $a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}}$.

Definition 3.1 (Generating series). Let $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ be finite sets of positive integers. The **generating series** associated to $[\mathcal{A}_0; \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n]$ is the formal power series in $\mathbb{Q}[[t]]$ defined by

$$[\mathcal{A}_0; \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n](t) = \mathcal{A}_0(t) + \cfrac{1}{\mathcal{A}_1(t) + \cfrac{1}{\mathcal{A}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t)}}}}.$$

Lemma 3.2 (Recursive generating series). Let $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ be finite sets of positive integers. Let $\mathcal{A}_0(t), \mathcal{A}_1(t), \mathcal{A}_2(t), \ldots, \mathcal{A}_n(t)$ be their generating series. Then

$$[\mathcal{A}_0; \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n](t) = \mathcal{A}_0(t) + \frac{1}{[\mathcal{A}_1; \mathcal{A}_2, \ldots, \mathcal{A}_n](t)}.$$

Proof. Let $[\mathcal{A}_1; \mathcal{A}_2, \ldots, \mathcal{A}_n](t) = \mathcal{A}_1(t) + \cfrac{1}{\mathcal{A}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t)}}}$. Then

$$[\mathcal{A}_0; \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n](t) = \mathcal{A}_0(t) + \cfrac{1}{\mathcal{A}_1(t) + \cfrac{1}{\mathcal{A}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t)}}}} = \mathcal{A}_0(t) + \frac{1}{[\mathcal{A}_1; \mathcal{A}_2, \ldots, \mathcal{A}_n](t)}.$$

□

Lemma 3.3 (Reversal). Let $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ be finite sets of positive integers. Let $\mathcal{A}_0(t), \mathcal{A}_1(t), \mathcal{A}_2(t), \ldots, \mathcal{A}_n(t)$ be their generating series. Then

$$[\mathcal{A}_0; \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n](t) = \mathcal{A}_0(t) + \frac{1}{[\mathcal{A}_n; \mathcal{A}_{n-1}, \ldots, \mathcal{A}_1](t)}.$$

Proof. Let $[\mathcal{A}_n; \mathcal{A}_{n-1}, \ldots, \mathcal{A}_1](t) = \mathcal{A}_n(t) + \cfrac{1}{\mathcal{A}_{n-1}(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_1(t)}}}$. Then

$$[\mathcal{A}_0; \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n](t) = \mathcal{A}_0(t) + \cfrac{1}{\mathcal{A}_1(t) + \cfrac{1}{\mathcal{A}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t)}}}} = \mathcal{A}_0(t) + \frac{1}{[\mathcal{A}_n; \mathcal{A}_{n-1}, \ldots, \mathcal{A}_1](t)}.$$

□

Lemma 3.4 (Product and sum formulas). Let $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ be finite sets of positive integers. Let $\mathcal{A}_0(t), \mathcal{A}_1(t), \mathcal{A}_2(t), \ldots, \mathcal{A}_n(t)$ be their generating series. Let $\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2, \ldots, \mathcal{B}_n$ be finite sets of positive integers. Let $\mathcal{B}_0(t), \mathcal{B}_1(t), \mathcal{B}_2(t), \ldots, \mathcal{B}_n(t)$ be their generating series. Then

$$[\mathcal{A}_0 \cdot \mathcal{B}_0; \mathcal{A}_1 \cdot \mathcal{B}_1, \mathcal{A}_2 \cdot \mathcal{B}_2, \ldots, \mathcal{A}_n \cdot \mathcal{B}_n](t) = \mathcal{A}_0(t)\,\mathcal{B}_0(t) + \frac{1}{[\mathcal{A}_1 \cdot \mathcal{B}_1; \mathcal{A}_2 \cdot \mathcal{B}_2, \ldots, \mathcal{A}_n \cdot \mathcal{B}_n](t)},$$

$$[\mathcal{A}_0 + \mathcal{B}_0; \mathcal{A}_1 + \mathcal{B}_1, \mathcal{A}_2 + \mathcal{B}_2, \ldots, \mathcal{A}_n + \mathcal{B}_n](t) = \mathcal{A}_0(t) + \mathcal{B}_0(t) + \frac{1}{[\mathcal{A}_1 + \mathcal{B}_1; \mathcal{A}_2 + \mathcal{B}_2, \ldots, \mathcal{A}_n + \mathcal{B}_n](t)}.$$

Proof. Let $[\mathcal{A}_1 \cdot \mathcal{B}_1; \mathcal{A}_2 \cdot \mathcal{B}_2, \ldots, \mathcal{A}_n \cdot \mathcal{B}_n](t) = \mathcal{A}_1(t)\,\mathcal{B}_1(t) + \cfrac{1}{\mathcal{A}_2(t)\,\mathcal{B}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t)\,\mathcal{B}_n(t)}}}$. Then

$$[\mathcal{A}_0 \cdot \mathcal{B}_0; \mathcal{A}_1 \cdot \mathcal{B}_1, \mathcal{A}_2 \cdot \mathcal{B}_2, \ldots, \mathcal{A}_n \cdot \mathcal{B}_n](t) = \mathcal{A}_0(t)\,\mathcal{B}_0(t) + \cfrac{1}{\mathcal{A}_1(t)\,\mathcal{B}_1(t) + \cfrac{1}{\mathcal{A}_2(t)\,\mathcal{B}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t)\,\mathcal{B}_n(t)}}}} = \mathcal{A}_0(t)\,\mathcal{B}_0(t) + \frac{1}{[\mathcal{A}_1 \cdot \mathcal{B}_1; \mathcal{A}_2 \cdot \mathcal{B}_2, \ldots, \mathcal{A}_n \cdot \mathcal{B}_n](t)}.$$

Let $[\mathcal{A}_1 + \mathcal{B}_1; \mathcal{A}_2 + \mathcal{B}_2, \ldots, \mathcal{A}_n + \mathcal{B}_n](t) = \mathcal{A}_1(t) + \mathcal{B}_1(t) + \cfrac{1}{\mathcal{A}_2(t) + \mathcal{B}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t) + \mathcal{B}_n(t)}}}$. Then

$$[\mathcal{A}_0 + \mathcal{B}_0; \mathcal{A}_1 + \mathcal{B}_1, \mathcal{A}_2 + \mathcal{B}_2, \ldots, \mathcal{A}_n + \mathcal{B}_n](t) = \mathcal{A}_0(t) + \mathcal{B}_0(t) + \cfrac{1}{\mathcal{A}_1(t) + \mathcal{B}_1(t) + \cfrac{1}{\mathcal{A}_2(t) + \mathcal{B}_2(t) + \cfrac{1}{\ddots + \cfrac{1}{\mathcal{A}_n(t) + \mathcal{B}_n(t)}}}} = \mathcal{A}_0(t) + \mathcal{B}_0(t) + \frac{1}{[\mathcal{A}_1 + \mathcal{B}_1; \mathcal{A}_2 + \mathcal{B}_2, \ldots, \mathcal{A}_n + \mathcal{B}_n](t)}.$$

□

Let Γ denote the galoisian group of $\mathbb{Q}$.

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Let Γ denote the galoisian group of $\mathbb{Q}$.

Lemma 1. Let Γ be a Kronecker-Weyl field. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{theorem}[Kronecker-Weyl, Lambert, Lobachevskii-Wiles]
Let $\Gamma$ be a Kronecker-Weyl field. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl.
\end{theorem}
```

Lemma 2. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 3. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 4. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 5. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 6. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 7. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 8. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 9. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 10. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 11. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 12. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 13. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
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Lemma 14. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 15. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 16. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 17. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 18. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 19. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 20. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 21. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 22. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 23. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 24. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 25. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 26. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 27. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Lemma 28. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.

Proof. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let Γ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then Γ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles. Therefore Γ is Cauchy-Weyl.

```
\begin{lemma}
Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Let $\Gamma$ be a Cauchy-Weyl, Lambert, Lobachevskii-Wiles vector. Then $\Gamma$ is Cauchy-Weyl, Lambert, Lobachevskii-Wiles.
\end{lemma}
```

Jurisdiction Decision - 2011

Decision Number: 2011-0-1

Decision Date: 2011-01-01

Decision Text: **The Hague Hold an Action Plan Office (jurisdiction) as a co-co, the Hague Municipalities, the Hague and Leiden to the action plan.**

2.2.1 Determine the domain and range of the function $f(x) = \sqrt{\frac{x+2}{x-3}}$ and sketch the graph of f.

2.2.2 Determine the domain and range of the function $f(x) = \sqrt{\frac{x^2-4}{x^2-9}}$ and sketch the graph of f.

2.2.3 Let $f(x) = \frac{x^2-4}{x-2}$ and $g(x) = x+2$.

a. Is $f(x) = g(x)$ for all x? Why or why not?

b. Graph f and g on the same axes.

2.2.4 Let $f(x) = \frac{x^2-4}{x-2}$ and $g(x) = x+2$. Is $f(x) = g(x)$ for all x? Why or why not?

2.2.5 Let $f(x) = \frac{x^2-4}{x-2}$ and $g(x) = x+2$. Is $f(x) = g(x)$ for all x? Why or why not?

2.2.6 Let $f(x) = \frac{x^2-4}{x-2}$ and $g(x) = x+2$. Is $f(x) = g(x)$ for all x? Why or why not?

```
[ ]: !pip install -q -U google-genai[local] ipykeyboard
```

```
[ ]: !pip install -q -U google-genai[local] ipykeyboard
!sudo apt-get -qq -y install -q -U -t bookworm -o Dpkg::Options::="--force-confdef" -o Dpkg::Options::="--force-dep
```

```
[ ]: %%file /usr/local/lib/python3.12/dist-packages/IPykeyboard/__init__.pyi
```

[ ]: `%%file /usr/local/lib/python3.12/dist-packages/IPykeyboard/__init__.pyi`

[illegible] [illegible] [illegible] [

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sym.Function('a')(i,j))
```

$$\begin{bmatrix} a(0, 0) & a(0, 1) \\ a(1, 0) & a(1, 1) \end{bmatrix}$$

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('a')(i,k)*Sym.Function('b')(k,j),(k,0,2)))
```

$$\begin{bmatrix} a(0, 0)b(0, 0) + a(0, 1)b(1, 0) & a(0, 0)b(0, 1) + a(0, 1)b(1, 1) \\ a(1, 0)b(0, 0) + a(1, 1)b(1, 0) & a(1, 0)b(0, 1) + a(1, 1)b(1, 1) \end{bmatrix}$$

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('a')(i,k)*Sym.Function('b')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('c')(i,k)*Sym.Function('d')(k,j),(k,0,2)))
```

$$\begin{bmatrix} a(0, 0)b(0, 0)c(0, 0) + a(0, 0)b(0, 1)c(1, 0) + a(0, 1)b(1, 0)c(0, 0) + a(0, 1)b(1, 1)c(1, 0) & a(0, 0)b(0, 0)c(0, 1) + a(0, 0)b(0, 1)c(1, 1) + a(0, 1)b(1, 0)c(0, 1) + a(0, 1)b(1, 1)c(1, 1) \\ a(1, 0)b(0, 0)c(0, 0) + a(1, 0)b(0, 1)c(1, 0) + a(1, 1)b(1, 0)c(0, 0) + a(1, 1)b(1, 1)c(1, 0) & a(1, 0)b(0, 0)c(0, 1) + a(1, 0)b(0, 1)c(1, 1) + a(1, 1)b(1, 0)c(0, 1) + a(1, 1)b(1, 1)c(1, 1) \end{bmatrix}$$

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('a')(i,k)*Sym.Function('b')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('c')(i,k)*Sym.Function('d')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('e')(i,k)*Sym.Function('f')(k,j),(k,0,2)))
```

$$\begin{bmatrix} a(0, 0)b(0, 0)c(0, 0)e(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0) & a(0, 0)b(0, 0)c(0, 0)f(0, 0) + a(0, 0)b(0, 0)c(0, 0)f(1, 0) + a(0, 0)b(0, 1)c(1, 0)f(0, 0) + a(0, 0)b(0, 1)c(1, 0)f(1, 0) + a(0, 1)b(1, 0)c(0, 0)f(0, 0) + a(0, 1)b(1, 0)c(0, 0)f(1, 0) + a(0, 1)b(1, 1)c(1, 0)f(0, 0) + a(0, 1)b(1, 1)c(1, 0)f(1, 0) \\ a(1, 0)b(0, 0)c(0, 0)e(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0) & a(1, 0)b(0, 0)c(0, 0)f(0, 0) + a(1, 0)b(0, 0)c(0, 0)f(1, 0) + a(1, 0)b(0, 1)c(1, 0)f(0, 0) + a(1, 0)b(0, 1)c(1, 0)f(1, 0) + a(1, 1)b(1, 0)c(0, 0)f(0, 0) + a(1, 1)b(1, 0)c(0, 0)f(1, 0) + a(1, 1)b(1, 1)c(1, 0)f(0, 0) + a(1, 1)b(1, 1)c(1, 0)f(1, 0) \end{bmatrix}$$

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('a')(i,k)*Sym.Function('b')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('c')(i,k)*Sym.Function('d')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('e')(i,k)*Sym.Function('f')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('g')(i,k)*Sym.Function('h')(k,j),(k,0,2)))
```

$$\begin{bmatrix} a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0) & a(0, 0)b(0, 0)c(0, 0)e(0, 0)h(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)h(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)h(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)h(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)h(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)h(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)h(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)h(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)h(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)h(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)h(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)h(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)h(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)h(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)h(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)h(1, 0) \\ a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0) & a(1, 0)b(0, 0)c(0, 0)e(0, 0)h(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(0, 0)h(1, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)h(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)h(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)h(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)h(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)h(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)h(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)h(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)h(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)h(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)h(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)h(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)h(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)h(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)h(1, 0) \end{bmatrix}$$

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('a')(i,k)*Sym.Function('b')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('c')(i,k)*Sym.Function('d')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('e')(i,k)*Sym.Function('f')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('g')(i,k)*Sym.Function('h')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('i')(i,k)*Sym.Function('j')(k,j),(k,0,2)))
```

$$\begin{bmatrix} a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)i(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)i(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)i(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)i(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)i(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)i(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)i(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)i(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)i(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)i(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)i(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)i(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)i(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)i(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)i(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)i(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0)i(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0)i(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0)i(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0)i(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)i(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)i(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)i(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)i(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)i(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)i(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)i(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)i(1, 0) & a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)j(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)j(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)j(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)j(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)j(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)j(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)j(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)j(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)j(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)j(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)j(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)j(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)j(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)j(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)j(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)j(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)j(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)j(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)j(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)j(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0)j(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0)j(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0)j(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0)j(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)j(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)j(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)j(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)j(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)j(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)j(1, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)j(0, 0) + a(0, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)j(1, 0) \\ a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0) + a(1, 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1)c(1, 0)e(0, 0)g(0, 0)i(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)i(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)i(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)i(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)i(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)i(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)i(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)i(1, 0) & a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)j(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)j(1, 0) + a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)j(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)j(1, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)j(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)j(1, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)j(0, 0) + a(1, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)j(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)j(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)j(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)j(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)j(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)j(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)j(1, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)j(0, 0) + a(1, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)j(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)j(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)j(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)j(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)j(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0)j(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)g(0, 0)j(1, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0)j(0, 0) + a(1, 1)b(1, 0)c(0, 0)e(1, 0)g(1, 0)j(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)j(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(0, 0)j(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)j(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(0, 0)g(1, 0)j(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)j(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(0, 0)j(1, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)j(0, 0) + a(1, 1)b(1, 1)c(1, 0)e(1, 0)g(1, 0)j(1, 0) \end{bmatrix}$$

```
Sym.clear(Sym.Matrix(2*, 2*))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('a')(i,k)*Sym.Function('b')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('c')(i,k)*Sym.Function('d')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('e')(i,k)*Sym.Function('f')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('g')(i,k)*Sym.Function('h')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('i')(i,k)*Sym.Function('j')(k,j),(k,0,2)))*Sym.Matrix(2, 2, lambda i,j:Sum(Sym.Function('k')(i,k)*Sym.Function('l')(k,j),(k,0,2)))
```

$$\begin{bmatrix} a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(0, 0)g(1, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(0, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 0)c(0, 0)e(1, 0)g(1, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(0, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(0, 0)g(1, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(0, 0)i(1, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)i(0, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)i(0, 0)k(1, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)i(1, 0)k(0, 0) + a(0, 0)b(0, 1)c(1, 0)e(1, 0)g(1, 0)i(1, 0)k(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0)k(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)i(0, 0)k(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0)k(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(0, 0)i(1, 0)k(1, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)i(0, 0)k(0, 0) + a(0, 1)b(1, 0)c(0, 0)e(0, 0)g(1, 0)i(0, 0)k(1, 0) + a(0, 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一种可调节的电子烟雾化器

[57]摘要

本发明涉及一种可调节的电子烟雾化器，包括：雾化器主体，设置在该主体内部的雾化芯；以及调节机构，该调节机构包括：第一调节环，设置在该雾化器主体的外侧，该第一调节环可沿该雾化器主体的轴向移动；第二调节环，设置在该第一调节环的内侧，该第二调节环可沿该雾化器主体的轴向移动；第一弹簧，设置在该第一调节环与该雾化器主体之间，该第一弹簧用于向该第一调节环施加朝向该雾化器主体的预紧力；第二弹簧，设置在该第二调节环与该第一调节环之间，该第二弹簧用于向该第二调节环施加朝向该第一调节环的预紧力；第一限位凸台，设置在该雾化器主体的外侧，该第一限位凸台用于限制该第一调节环沿该雾化器主体的轴向移动的最大行程；第二限位凸台，设置在该第一调节环的外侧，该第二限位凸台用于限制该第二调节环沿该第一调节环的轴向移动的最大行程；第一卡槽，设置在该第一调节环的内侧，该第一卡槽用于与该第一限位凸台配合；第二卡槽，设置在该第二调节环的内侧，该第二卡槽用于与该第二限位凸台配合；第一卡扣，设置在该第一调节环的外侧，该第一卡扣用于与该第二卡槽配合；第二卡扣，设置在该第二调节环的外侧，该第二卡扣用于与该第一卡槽配合；第一卡槽，设置在该第一调节环的外侧，该第一卡槽用于与该第二卡扣配合；第二卡槽，设置在该第二调节环的外侧，该第二卡槽用于与该第一卡扣配合；第一卡扣，设置在该第一调节环的内侧，该第一卡扣用于与该第二限位凸台配合；第二卡扣，设置在该第二调节环的内侧，该第二卡扣用于与该第一限位凸台配合；第一卡槽，设置在该第一调节环的内侧，该第一卡槽用于与该第二卡扣配合；第二卡槽，设置在该第二调节环的内侧，该第二卡槽用于与该第一卡扣

$$\frac{\partial}{\partial t}\int_{\Omega}\rho\,\mathbf{v}\,dV = \int_{\Omega}\rho\,\mathbf{g}\,dV + \int_{\Omega}\nabla\cdot\mathbf{T}\,dV$$

Image processing is the manipulation of digital images by computer. The modern digital computer was not designed with image processing in mind, and consequently the manipulation of even the simplest digital image by computer can be a cumbersome and time-consuming exercise. However, the ever-increasing speed and power of computers, coupled with the development of new software and hardware for the processing of digital images, has ensured that what was once the domain of a few specialists with access to expensive and specialized equipment is now accessible to anyone who has a personal computer and a digital camera. Indeed, the ubiquitous digital camera, which is now a feature of most mobile phones, has ushered in a new era of digital image processing, with billions of digital images being created and processed every day.

1.1 Introduction

Image processing is the manipulation of digital images by computer. The modern digital computer was not designed with image processing in mind, and consequently the manipulation of even the simplest digital image by computer can be a cumbersome and time-consuming exercise. However, the ever-increasing speed and power of computers, coupled with the development of new software and hardware for the processing of digital images, has ensured that what was once the domain of a few specialists with access to expensive and specialized equipment is now accessible to anyone who has a personal computer and a digital camera. Indeed, the ubiquitous digital camera, which is now a feature of most mobile phones, has ushered in a new era of digital image processing, with billions of digital images being created and processed every day.

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Fibonacci Sequence and Golden Ratio Analysis

Introduction

The Fibonacci sequence is a sequence of numbers in which each number is the sum of the two preceding numbers. Formally, the **Fibonacci sequence** $\{F_n\}_{n=0}^{\infty}$ is defined by the **recurrence relation**

$$F_n = F_{n-1} + F_{n-2}$$

Theorem

Theorem

Let $\{F_n\}_{n=0}^{\infty}$ be the Fibonacci sequence. Then

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \frac{1+\sqrt{5}}{2}.$$

Proof

Let $\{F_n\}_{n=0}^{\infty}$ be the Fibonacci sequence.

Let $\{a_n\}_{n=0}^{\infty}$ be the sequence defined by $a_n = \frac{F_{n+1}}{F_n}$ for all $n \in \mathbb{N}$.

Let $\{b_n\}_{n=0}^{\infty}$ be the sequence defined by $b_n = \frac{F_n}{F_{n-1}}$ for all $n \in \mathbb{N}$.

Let $\{c_n\}_{n=0}^{\infty}$ be the sequence defined by $c_n = a_n - b_n$ for all $n \in \mathbb{N}$.

Let $\{d_n\}_{n=0}^{\infty}$ be the sequence defined by $d_n = \frac{c_n}{a_n}$ for all $n \in \mathbb{N}$.

Let $\{e_n\}_{n=0}^{\infty}$ be the sequence defined by $e_n = \frac{d_n}{c_n}$ for all $n \in \mathbb{N}$.

Let $\{f_n\}_{n=0}^{\infty}$ be the sequence defined by $f_n = \frac{e_n}{d_n}$ for all $n \in \mathbb{N}$.

Let $\{g_n\}_{n=0}^{\infty}$ be the sequence defined by $g_n = \frac{f_n}{e_n}$ for all $n \in \mathbb{N}$.

Let $\{h_n\}_{n=0}^{\infty}$ be the sequence defined by $h_n = \frac{g_n}{f_n}$ for all $n \in \mathbb{N}$.

Let $\{i_n\}_{n=0}^{\infty}$ be the sequence defined by $i_n = \frac{h_n}{g_n}$ for all $n \in \mathbb{N}$.

Let $\{j_n\}_{n=0}^{\infty}$ be the sequence defined by $j_n = \frac{i_n}{h_n}$ for all $n \in \mathbb{N}$.

Let $\{k_n\}_{n=0}^{\infty}$ be the sequence defined by $k_n = \frac{j_n}{i_n}$ for all $n \in \mathbb{N}$.

Let $\{l_n\}_{n=0}^{\infty}$ be the sequence defined by $l_n = \frac{k_n}{j_n}$ for all $n \in \mathbb{N}$.

Let $\{m_n\}_{n=0}^{\infty}$ be the sequence defined by $m_n = \frac{l_n}{k_n}$ for all $n \in \mathbb{N}$.

Let $\{n_n\}_{n=0}^{\infty}$ be the sequence defined by $n_n = \frac{m_n}{l_n}$ for all $n \in \mathbb{N}$.

Let $\{o_n\}_{n=0}^{\infty}$ be the sequence defined by $o_n = \frac{n_n}{m_n}$ for all $n \in \mathbb{N}$.

Let $\{p_n\}_{n=0}^{\infty}$ be the sequence defined by $p_n = \frac{o_n}{n_n}$ for all $n \in \mathbb{N}$.

Let $\{q_n\}_{n=0}^{\infty}$ be the sequence defined by $q_n = \frac{p_n}{o_n}$ for all $n \in \mathbb{N}$.

Let $\{r_n\}_{n=0}^{\infty}$ be the sequence defined by $r_n = \frac{q_n}{p_n}$ for all $n \in \mathbb{N}$.

Let $\{s_n\}_{n=0}^{\infty}$ be the sequence defined by $s_n = \frac{r_n}{q_n}$ for all $n \in \mathbb{N}$.

Let $\{t_n\}_{n=0}^{\infty}$ be the sequence defined by $t_n = \frac{s_n}{r_n}$ for all $n \in \mathbb{N}$.

Let $\{u_n\}_{n=0}^{\infty}$ be the sequence defined by $u_n = \frac{t_n}{s_n}$ for all $n \in \mathbb{N}$.

Let $\{v_n\}_{n=0}^{\infty}$ be the sequence defined by $v_n = \frac{u_n}{t_n}$ for all $n \in \mathbb{N}$.

Let $\{w_n\}_{n=0}^{\infty}$ be the sequence defined by $w_n = \frac{v_n}{u_n}$ for all $n \in \mathbb{N}$.

Let $\{x_n\}_{n=0}^{\infty}$ be the sequence defined by $x_n = \frac{w_n}{v_n}$ for all $n \in \mathbb{N}$.

Let $\{y_n\}_{n=0}^{\infty}$ be the sequence defined by $y_n = \frac{x_n}{w_n}$ for all $n \in \mathbb{N}$.

Let $\{z_n\}_{n=0}^{\infty}$ be the sequence defined by $z_n = \frac{y_n}{x_n}$ for all $n \in \mathbb{N}$.

Let $\{a_{1,n}\}_{n=0}^{\infty}$ be the sequence defined by $a_{1,n} = \frac{z_n}{y_n}$ for all $n \in \mathbb{N}$.

Let $\{a_{2,n}\}_{n=0}^{\infty}$ be the sequence defined by $a_{2,n} = \frac{a_{1,n}}{z_n}$ for all $n \in \mathbb{N}$.

Let $\{a_{3,n}\}_{n=0}^{\infty}$ be the sequence defined by $a_{3,n} = \frac{a_{2,n}}{a_{1,n}}$ for all $n \in \mathbb{N}$.

Let $\{a_{4,n}\}_{n=0}^{\infty}$ be the sequence defined by $a_{4,n} = \frac{a_{3,n}}{a_{2,n}}$ for all $n \in \mathbb{N}$.

Let $\{a_{5,n}\}_{n=0}^{\infty}$ be the sequence defined by $a_{5,n} = \frac{a_{4,n}}{a_{3,n}}$ for all $n \in \mathbb{N}$.

Let

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Ecosystem services and biodiversity in a landscape context

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Abstract

1. Ecosystem services are the benefits people derive from ecosystems. Although a growing literature has begun to quantify the value of ecosystem services, little attention has been paid to the role of biodiversity in the provision of these services. Here we review the evidence for the effects of biodiversity on ecosystem services, and we discuss how landscape structure may influence the provision of ecosystem services.

2. We find that biodiversity has a positive effect on the provision of ecosystem services, and that landscape structure can have a strong effect on the provision of ecosystem services. We also find that the effects of biodiversity on ecosystem services may be mediated by landscape structure, and that landscape structure may be a proxy for biodiversity in the provision of ecosystem services.

3. *Synthesis and applications.* Our review suggests that biodiversity is a key component of ecosystem services, and that landscape structure may be a useful proxy for biodiversity in the provision of ecosystem services. We also suggest that landscape structure may be a useful tool for the management of ecosystem services.

Keywords

ecosystem services, biodiversity, landscape structure, landscape ecology, ecosystem services, biodiversity, landscape structure, landscape ecology, ecosystem services, biodiversity, landscape structure, landscape ecology,

1. Which of the following is a true statement about the Pythagorean theorem?

A. The Pythagorean theorem states that $a^2 + b^2 = c^2$ for any right triangle.

B. The Pythagorean theorem states that $a^2 + b^2 = c^2$ for any triangle.

C. The Pythagorean theorem states that $a^2 + b^2 = c^2$ for any isosceles triangle.

D. The Pythagorean theorem states that $a^2 + b^2 = c^2$ for any equilateral triangle.

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Chapter 1. Mathematical Foundations for Quantum Computing

1.1. Linear Algebra: Vectors and Matrices

1.1.1. Vectors and Their Properties

Definition 1.1.1 (Vector Space). A vector space V over the field $\mathbb{C}$ satisfies:

$$\forall v, w \in V, \forall \alpha \in \mathbb{C}: \quad v + w \in V, \quad \alpha v \in V \tag{1.1.1.1}$$

Definition 1.1.2 (Inner Product). The inner product $\langle \cdot, \cdot \rangle$ satisfies:

$$\langle v, w \rangle = \overline{\langle w, v \rangle}, \quad \langle v, v \rangle \geq 0, \quad \langle v, v \rangle = 0 \iff v = 0 \tag{1.1.1.2}$$

Definition 1.1.3 (Norm). The norm induced by the inner product:

$$\|v\| = \sqrt{\langle v, v \rangle} \tag{1.1.1.3}$$

Definition 1.1.4 (Orthogonal Projection). The projection of v onto w:

$$P_w(v) = \frac{\langle w, v \rangle}{\langle w, w \rangle} w \tag{1.1.1.4}$$

1.1.2. Matrices and Their Properties

Definition 1.1.5 (Matrix Multiplication). For $A \in \mathbb{C}^{m \times n}$, $B \in \mathbb{C}^{n \times p}$:

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj} \tag{1.1.1.5}$$

Definition 1.1.6 (Hermitian Conjugate). The Hermitian conjugate of A:

$$A^\dagger = \overline{A^T} \tag{1.1.1.6}$$

Definition 1.1.7 (Trace). The trace of A:

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} A_{ii} \tag{1.1.1.7}$$

1.1.3. Eigenvalues and Eigenvectors

Definition 1.1.8 (Eigenvalue Equation). For $A \in \mathbb{C}^{n \times n}$:

$$Av = \lambda v \tag{1.1.1.8}$$

Definition 1.1.9 (Characteristic Polynomial). The characteristic polynomial of A:

$$p_A(\lambda) = \det(A - \lambda I) \tag{1.1.1.9}$$

1.1.4. Tensor Products and Multi-Qubit Systems

Definition 1.1.10 (Tensor Product). For $A \in \mathbb{C}^{m \times n}$, $B \in \mathbb{C}^{p \times q}$:

$$(A \otimes B)_{(i,p),(j,q)} = A_{ij} B_{pq} \tag{1.1.1.10}$$

1.1.5. Unitary Matrices and Quantum Gates

Definition 1.1.11 (Unitary Matrix). A matrix U satisfies:

$$U^\dagger U = I \tag{1.1.1.11}$$

Definition 1.1.12 (Pauli Matrices). The Pauli matrices are:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{1.1.1.12}$$

1.1.6. Hermitian Matrices and Observables

Definition 1.1.13 (Hermitian Matrix). A matrix satisfies:

$$A = A^\dagger \tag{1.1.1.13}$$

1.1.7. Spectral Decomposition

Theorem 1.1.14 (Spectral Decomposition). Any Hermitian matrix A can be decomposed as:

$$A = \sum_{i=1}^{n} \lambda_i |v_i\rangle\langle v_i| \tag{1.1.1.14}$$

1.1.8. Trace and Its Properties

Theorem 1.1.15 (Trace Properties). The trace satisfies:

$$\mathrm{Tr}(AB) = \mathrm{Tr}(BA), \quad \mathrm{Tr}(A \otimes B) = \mathrm{Tr}(A)\mathrm{Tr}(B) \tag{1.1.1.15}$$

1.1.9. Matrix Exponential

Definition 1.1.16 (Matrix Exponential). The matrix exponential of A:

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!} \tag{1.1.1.16}$$

1.1.10. Pauli Matrices

Definition 1.1.17 (Pauli Matrices). The Pauli matrices satisfy:

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I, \quad \sigma_x \sigma_y = i\sigma_z, \quad \sigma_y \sigma_z = i\sigma_x, \quad \sigma_z \sigma_x = i\sigma_y \tag{1.1.1.17}$$

1.1.11. Summary

1.2. Probability Theory and Stochastic Processes

1.2.1. Probability Spaces

Definition 1.2.1 (Probability Space). A probability space $(\Omega, \mathcal{F}, P)$ consists of:

- Ω: sample space
- $\mathcal{F}$: σ-algebra of events
- P: probability measure satisfying:

$$P(\emptyset) = 0, \quad P(\Omega) = 1, \quad P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \tag{1.2.1.1}$$

1.2.2. Random Variables

Definition 1.2.2 (Random Variable). A random variable X is a measurable function from Ω to $\mathbb{R}$:

$$X^{-1}(B) \in \mathcal{F} \quad \forall B \in \mathcal{B}(\mathbb{R}) \tag{1.2.1.2}$$

1.2.3. Expectation and Variance

Definition 1.2.3 (Expectation). The expectation of X:

$$E[X] = \int_\Omega X \, dP \tag{1.2.1.3}$$

Definition 1.2.4 (Variance). The variance of X:

$$\mathrm{Var}(X) = E[(X - E[X])^2] \tag{1.2.1.4}$$

1.2.4. Conditional Probability and Bayes' Theorem

Definition 1.2.5 (Conditional Probability). The conditional probability of A given B:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{1.2.1.5}$$

Theorem 1.2.6 (Bayes' Theorem). Bayes' theorem states:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{1.2.1.6}$$

1.2.5. Stochastic Processes

Definition 1.2.7 (Stochastic Process). A stochastic process is a family $\{X_t\}_{t \in T}$ of random variables indexed by T.

1.2.6. Markov Chains

Definition 1.2.8 (Markov Chain). A stochastic process satisfies the Markov property if:

$$P(X_{t+1} = j | X_t = i, X_{t-1}, \ldots) = P(X_{t+1} = j | X_t = i) \tag{1.2.1.8}$$

1.2.7. Martingales

Definition 1.2.9 (Martingale). A stochastic process satisfies:

$$E[X_{t+1} | \mathcal{F}_t] = X_t \tag{1.2.1.9}$$

1.2.8. Brownian Motion

Definition 1.2.10 (Brownian Motion). A stochastic process satisfies:

$$W_0 = 0, \quad W_t - W_s \sim N(0, t-s), \quad W_t \text{ has independent increments} \tag{1.2.1.10}$$

1.2.9. Stochastic Integrals

Definition 1.2.11 (Stochastic Integral). The stochastic integral:

$$\int_0^t f(s) \, dW_s \tag{1.2.1.11}$$

1.2.10. Stochastic Differential Equations

Definition 1.2.12 (Stochastic Differential Equation). A stochastic differential equation:

$$dX_t = \mu(X_t, t) \, dt + \sigma(X_t, t) \, dW_t \tag{1.2.1.12}$$

1.2.11. Summary

1.3. Differential Equations

1.3.1. Ordinary Differential Equations

Definition 1.3.1 (Ordinary Differential Equation). An ordinary differential equation:

$$F(x, y, y', \ldots, y^{(n)}) = 0 \tag{1.3.1.1}$$

1.3.2. Partial Differential Equations

Definition 1.3.2 (Partial Differential Equation). A partial differential equation:

$$F(x_1, \ldots, x_n, u, \partial u, \ldots, \partial^k u) = 0 \tag{1.3.1.2}$$

1.3.3. Boundary Value Problems

Definition 1.3.3 (Boundary Value Problem). A boundary value problem:

$$\begin{cases} F(x, y, y', \ldots, y^{(n)}) = 0 \\ B(y(a), y(b), \ldots) = 0 \end{cases} \tag{1.3.1.3}$$

1.3.4. Initial Value Problems

Definition 1.3.4 (Initial Value Problem). An initial value problem:

$$\begin{cases} F(x, y, y', \ldots, y^{(n)}) = 0 \\ y(x_0) = y_0, \ldots, y^{(n-1)}(x_0) = y_{n-1} \end{cases} \tag{1.3.1.4}$$

1.3.5. Numerical Methods

1.3.6. Stability Analysis

1.3.7. Summary

1.4. Optimization Theory

1.4.1. Unconstrained Optimization

Definition 1.4.1 (Unconstrained Optimization). Unconstrained optimization:

$$\min_{x \in \mathbb{R}^n} f(x) \tag{1.4.1.1}$$

1.4.2. Constrained Optimization

Definition 1.4.2 (Constrained Optimization). Constrained optimization:

$$\begin{cases} \min f(x) \\ g_i(x) \leq 0, \quad h_j(x) = 0 \end{cases} \tag{1.4.1.2}$$

1.4.3. Lagrange Multipliers

Definition 1.4.3 (Lagrange Multipliers). The Lagrangian:

$$L(x, \lambda, \mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x) \tag{1.4.1.3}$$

1.4.4. Convex Optimization

Definition 1.4.4 (Convex Optimization). Convex optimization:

$$\begin{cases} \min f(x) \\ g_i(x) \leq 0, \quad h_j(x) = 0 \end{cases} \tag{1.4.1.4}$$

1.4.5. Gradient Descent

Definition 1.4.5 (Gradient Descent). Gradient descent:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k) \tag{1.4.1.5}$$

1.4.6. Newton's Method

Definition 1.4.6 (Newton's Method). Newton's method:

$$x_{k+1} = x_k - H(x_k)^{-1} \nabla f(x_k) \tag{1.4.1.6}$$

1.4.7. Quasi-Newton Methods

Definition 1.4.7 (Quasi-Newton Methods). Quasi-Newton methods:

$$x_{k+1} = x_k - B_k^{-1} \nabla f(x_k) \tag{1.4.1.7}$$

1.4.8. Stochastic Gradient Descent

Definition 1.4.8 (Stochastic Gradient Descent). Stochastic gradient descent:

$$x_{k+1} = x_k - \alpha_k \nabla f_{i_k}(x_k) \tag{1.4.1.8}$$

1.4.9. Summary

1.5. Information Theory

1.5.1. Entropy

Definition 1.5.1 (Entropy). The entropy of a random variable X:

$$H(X) = -\sum_x p(x) \log p(x) \tag{1.5.1.1}$$

1.5.2. Mutual Information

Definition 1.5.2 (Mutual Information). The mutual information between X and Y:

$$I(X; Y) = H(X) - H(X|Y) \tag{1.5.1.2}$$

1.5.3. Relative Entropy

Definition 1.5.3 (Relative Entropy). The relative entropy:

$$D(p \| q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \tag{1.5.1.3}$$

1.5.4. Quantum Entropy

Definition 1.5.4 (Quantum Entropy). The quantum entropy:

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho) \tag{1.5.1.4}$$

1.5.5. Summary

1.6. Complexity Theory

1.6.1. Time Complexity

Definition 1.6.1 (Time Complexity). The time complexity:

$$T(n) = \max_{x: |x| = n} \text{time}(x) \tag{1.6.1.1}$$

1.6.2. Space Complexity

Definition 1.6.2 (Space Complexity). The space complexity:

$$S(n) = \max_{x: |x| = n} \text{space}(x) \tag{1.6.1.2}$$

1.6.3. Complexity Classes

Definition 1.6.3 (Complexity Classes). Complexity classes:

$$\mathrm{P}, \quad \mathrm{NP}, \quad \mathrm{BPP}, \quad \mathrm{BQP} \tag{1.6.1.3}$$

1.6.4. Quantum Complexity Classes

Definition 1.6.4 (Quantum Complexity Classes). Quantum complexity classes:

$$\mathrm{BQP}, \quad \mathrm{QMA}, \quad \mathrm{QCD} \tag{1.6.1.4}$$

1.6.5. Summary

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Abstract

Vitamin D

Cardiovascular

Chronic kidney disease

Hemodialysis

Mortality

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Natural numbers are infinite sets. Let $\mathbb{N}$ denote the set of natural numbers. Let $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ denote the set of natural numbers with zero.

Let $\mathbb{Z}$ denote the set of integers. Let $\mathbb{Q}$ denote the set of rational numbers. Let $\mathbb{R}$ denote the set of real numbers. Let $\mathbb{C}$ denote the set of complex numbers.

Let $\mathbb{P}$ denote the set of prime numbers. Let $\mathbb{E}$ denote the set of even numbers. Let $\mathbb{O}$ denote the set of odd numbers. Let $\mathbb{F}$ denote the set of Fibonacci numbers. Let $\mathbb{L}$ denote the set of Lucas numbers. Let $\mathbb{T}$ denote the set of triangular numbers. Let $\mathbb{S}$ denote the set of square numbers. Let $\mathbb{C}$ denote the set of cube numbers. Let $\mathbb{H}$ denote the set of hexagonal numbers. Let $\mathbb{P}_5$ denote the set of pentagonal numbers. Let $\mathbb{P}_7$ denote the set of heptagonal numbers. Let $\mathbb{O}_8$ denote the set of octagonal numbers. Let $\mathbb{N}_9$ denote the set of nonagonal numbers. Let $\mathbb{D}_{10}$ denote the set of decagonal numbers. Let $\mathbb{D}_{12}$ denote the set of dodecagonal numbers. Let $\mathbb{T}_{20}$ denote the set of icosagonal numbers. Let $\mathbb{T}_{30}$ denote the set of triacontagonal numbers. Let $\mathbb{T}_{40}$ denote the set of tetracontagonal numbers. 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Example 1.1.1. Consider a circular loop of radius a in the x-y plane, centered at the origin, and a long straight wire parallel to the z-axis, passing through the point $(x, y) = (b, 0)$ with $b > a$. The magnetic field due to the wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$, where $r = \sqrt{(x-b)^2 + y^2}$ is the distance from the wire to the point (x, y). Suppose the current I varies with time as $I(t) = I_0 \cos(\omega t)$. Find the induced emf in the loop.

Solution. Let $\mathbf{B}_{\text{wire}}$ denote the magnetic field due to the wire. Since $\mathbf{B}_{\text{wire}}$ points in the $\hat{\boldsymbol{\phi}}$ direction, it points in the $\hat{\mathbf{z}}$ direction everywhere on the loop. Therefore, the flux through the loop is

$$\Phi_B = \int_S \mathbf{B}_{\text{wire}} \cdot d\mathbf{a} = \int_S B_{\text{wire}}(x, y)\, da = \int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.1}$$

Since $I(t) = I_0 \cos(\omega t)$, the flux is a function of time. Differentiating with respect to time, we obtain the induced emf, which is given by Faraday's Law of Induction:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.2}$$

Since the loop is stationary, the integration region S does not depend on t, so we can pull the derivative inside the integral:

$$\mathcal{E} = -\int_S \frac{\mu_0}{2\pi \sqrt{(x-b)^2 + y^2}} \frac{dI(t)}{dt}\, da \tag{1.1.1.3}$$

Substituting $\frac{dI(t)}{dt} = -I_0 \omega \sin(\omega t)$, we obtain

$$\mathcal{E} = \int_S \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.4}$$

Since the integrand is independent of x and y, we can evaluate the integral by multiplying the integrand by the area of the loop, πa^2:

$$\mathcal{E} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}} \pi a^2 \tag{1.1.1.5}$$

Simplifying, we obtain

$$\mathcal{E} = \frac{\mu_0 I_0 \omega a^2 \sin(\omega t)}{2 \sqrt{(x-b)^2 + y^2}} \tag{1.1.1.6}$$

Exercise 1.1.1. Consider a circular loop of radius a in the x-y plane, centered at the origin, and a long straight wire parallel to the z-axis, passing through the point $(x, y) = (b, 0)$ with $b > a$. The magnetic field due to the wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$, where $r = \sqrt{(x-b)^2 + y^2}$ is the distance from the wire to the point (x, y). Suppose the current I varies with time as $I(t) = I_0 \cos(\omega t)$. Find the induced emf in the loop.

Solution. Let $\mathbf{B}_{\text{wire}}$ denote the magnetic field due to the wire. Since $\mathbf{B}_{\text{wire}}$ points in the $\hat{\boldsymbol{\phi}}$ direction, it points in the $\hat{\mathbf{z}}$ direction everywhere on the loop. Therefore, the flux through the loop is

$$\Phi_B = \int_S \mathbf{B}_{\text{wire}} \cdot d\mathbf{a} = \int_S B_{\text{wire}}(x, y)\, da = \int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.7}$$

Since $I(t) = I_0 \cos(\omega t)$, the flux is a function of time. Differentiating with respect to time, we obtain the induced emf, which is given by Faraday's Law of Induction:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.8}$$

Since the loop is stationary, the integration region S does not depend on t, so we can pull the derivative inside the integral:

$$\mathcal{E} = -\int_S \frac{\mu_0}{2\pi \sqrt{(x-b)^2 + y^2}} \frac{dI(t)}{dt}\, da \tag{1.1.1.9}$$

Substituting $\frac{dI(t)}{dt} = -I_0 \omega \sin(\omega t)$, we obtain

$$\mathcal{E} = \int_S \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.10}$$

Since the integrand is independent of x and y, we can evaluate the integral by multiplying the integrand by the area of the loop, πa^2:

$$\mathcal{E} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}} \pi a^2 \tag{1.1.1.11}$$

Simplifying, we obtain

$$\mathcal{E} = \frac{\mu_0 I_0 \omega a^2 \sin(\omega t)}{2 \sqrt{(x-b)^2 + y^2}} \tag{1.1.1.12}$$

Example 1.1.2. Consider a circular loop of radius a in the x-y plane, centered at the origin, and a long straight wire parallel to the z-axis, passing through the point $(x, y) = (b, 0)$ with $b > a$. The magnetic field due to the wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$, where $r = \sqrt{(x-b)^2 + y^2}$ is the distance from the wire to the point (x, y). Suppose the current I varies with time as $I(t) = I_0 \cos(\omega t)$. Find the induced emf in the loop.

Solution. Let $\mathbf{B}_{\text{wire}}$ denote the magnetic field due to the wire. Since $\mathbf{B}_{\text{wire}}$ points in the $\hat{\boldsymbol{\phi}}$ direction, it points in the $\hat{\mathbf{z}}$ direction everywhere on the loop. Therefore, the flux through the loop is

$$\Phi_B = \int_S \mathbf{B}_{\text{wire}} \cdot d\mathbf{a} = \int_S B_{\text{wire}}(x, y)\, da = \int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.13}$$

Since $I(t) = I_0 \cos(\omega t)$, the flux is a function of time. Differentiating with respect to time, we obtain the induced emf, which is given by Faraday's Law of Induction:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.14}$$

Since the loop is stationary, the integration region S does not depend on t, so we can pull the derivative inside the integral:

$$\mathcal{E} = -\int_S \frac{\mu_0}{2\pi \sqrt{(x-b)^2 + y^2}} \frac{dI(t)}{dt}\, da \tag{1.1.1.15}$$

Substituting $\frac{dI(t)}{dt} = -I_0 \omega \sin(\omega t)$, we obtain

$$\mathcal{E} = \int_S \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.16}$$

Since the integrand is independent of x and y, we can evaluate the integral by multiplying the integrand by the area of the loop, πa^2:

$$\mathcal{E} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}} \pi a^2 \tag{1.1.1.17}$$

Simplifying, we obtain

$$\mathcal{E} = \frac{\mu_0 I_0 \omega a^2 \sin(\omega t)}{2 \sqrt{(x-b)^2 + y^2}} \tag{1.1.1.18}$$

Exercise 1.1.2. Consider a circular loop of radius a in the x-y plane, centered at the origin, and a long straight wire parallel to the z-axis, passing through the point $(x, y) = (b, 0)$ with $b > a$. The magnetic field due to the wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$, where $r = \sqrt{(x-b)^2 + y^2}$ is the distance from the wire to the point (x, y). Suppose the current I varies with time as $I(t) = I_0 \cos(\omega t)$. Find the induced emf in the loop.

Solution. Let $\mathbf{B}_{\text{wire}}$ denote the magnetic field due to the wire. Since $\mathbf{B}_{\text{wire}}$ points in the $\hat{\boldsymbol{\phi}}$ direction, it points in the $\hat{\mathbf{z}}$ direction everywhere on the loop. Therefore, the flux through the loop is

$$\Phi_B = \int_S \mathbf{B}_{\text{wire}} \cdot d\mathbf{a} = \int_S B_{\text{wire}}(x, y)\, da = \int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.19}$$

Since $I(t) = I_0 \cos(\omega t)$, the flux is a function of time. Differentiating with respect to time, we obtain the induced emf, which is given by Faraday's Law of Induction:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.20}$$

Since the loop is stationary, the integration region S does not depend on t, so we can pull the derivative inside the integral:

$$\mathcal{E} = -\int_S \frac{\mu_0}{2\pi \sqrt{(x-b)^2 + y^2}} \frac{dI(t)}{dt}\, da \tag{1.1.1.21}$$

Substituting $\frac{dI(t)}{dt} = -I_0 \omega \sin(\omega t)$, we obtain

$$\mathcal{E} = \int_S \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.22}$$

Since the integrand is independent of x and y, we can evaluate the integral by multiplying the integrand by the area of the loop, πa^2:

$$\mathcal{E} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}} \pi a^2 \tag{1.1.1.23}$$

Simplifying, we obtain

$$\mathcal{E} = \frac{\mu_0 I_0 \omega a^2 \sin(\omega t)}{2 \sqrt{(x-b)^2 + y^2}} \tag{1.1.1.24}$$

Example 1.1.3. Consider a circular loop of radius a in the x-y plane, centered at the origin, and a long straight wire parallel to the z-axis, passing through the point $(x, y) = (b, 0)$ with $b > a$. The magnetic field due to the wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$, where $r = \sqrt{(x-b)^2 + y^2}$ is the distance from the wire to the point (x, y). Suppose the current I varies with time as $I(t) = I_0 \cos(\omega t)$. Find the induced emf in the loop.

Solution. Let $\mathbf{B}_{\text{wire}}$ denote the magnetic field due to the wire. Since $\mathbf{B}_{\text{wire}}$ points in the $\hat{\boldsymbol{\phi}}$ direction, it points in the $\hat{\mathbf{z}}$ direction everywhere on the loop. Therefore, the flux through the loop is

$$\Phi_B = \int_S \mathbf{B}_{\text{wire}} \cdot d\mathbf{a} = \int_S B_{\text{wire}}(x, y)\, da = \int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.25}$$

Since $I(t) = I_0 \cos(\omega t)$, the flux is a function of time. Differentiating with respect to time, we obtain the induced emf, which is given by Faraday's Law of Induction:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.26}$$

Since the loop is stationary, the integration region S does not depend on t, so we can pull the derivative inside the integral:

$$\mathcal{E} = -\int_S \frac{\mu_0}{2\pi \sqrt{(x-b)^2 + y^2}} \frac{dI(t)}{dt}\, da \tag{1.1.1.27}$$

Substituting $\frac{dI(t)}{dt} = -I_0 \omega \sin(\omega t)$, we obtain

$$\mathcal{E} = \int_S \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.28}$$

Since the integrand is independent of x and y, we can evaluate the integral by multiplying the integrand by the area of the loop, πa^2:

$$\mathcal{E} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}} \pi a^2 \tag{1.1.1.29}$$

Simplifying, we obtain

$$\mathcal{E} = \frac{\mu_0 I_0 \omega a^2 \sin(\omega t)}{2 \sqrt{(x-b)^2 + y^2}} \tag{1.1.1.30}$$

Exercise 1.1.3. Consider a circular loop of radius a in the x-y plane, centered at the origin, and a long straight wire parallel to the z-axis, passing through the point $(x, y) = (b, 0)$ with $b > a$. The magnetic field due to the wire is $\mathbf{B} = \frac{\mu_0 I}{2\pi r}\hat{\boldsymbol{\phi}}$, where $r = \sqrt{(x-b)^2 + y^2}$ is the distance from the wire to the point (x, y). Suppose the current I varies with time as $I(t) = I_0 \cos(\omega t)$. Find the induced emf in the loop.

Solution. Let $\mathbf{B}_{\text{wire}}$ denote the magnetic field due to the wire. Since $\mathbf{B}_{\text{wire}}$ points in the $\hat{\boldsymbol{\phi}}$ direction, it points in the $\hat{\mathbf{z}}$ direction everywhere on the loop. Therefore, the flux through the loop is

$$\Phi_B = \int_S \mathbf{B}_{\text{wire}} \cdot d\mathbf{a} = \int_S B_{\text{wire}}(x, y)\, da = \int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.31}$$

Since $I(t) = I_0 \cos(\omega t)$, the flux is a function of time. Differentiating with respect to time, we obtain the induced emf, which is given by Faraday's Law of Induction:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int_S \frac{\mu_0 I(t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.32}$$

Since the loop is stationary, the integration region S does not depend on t, so we can pull the derivative inside the integral:

$$\mathcal{E} = -\int_S \frac{\mu_0}{2\pi \sqrt{(x-b)^2 + y^2}} \frac{dI(t)}{dt}\, da \tag{1.1.1.33}$$

Substituting $\frac{dI(t)}{dt} = -I_0 \omega \sin(\omega t)$, we obtain

$$\mathcal{E} = \int_S \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}}\, da \tag{1.1.1.34}$$

Since the integrand is independent of x and y, we can evaluate the integral by multiplying the integrand by the area of the loop, πa^2:

$$\mathcal{E} = \frac{\mu_0 I_0 \omega \sin(\omega t)}{2\pi \sqrt{(x-b)^2 + y^2}} \pi a^2 \tag{1.1.1.35}$$

Simplifying, we obtain

$$\mathcal{E} = \frac{\mu_0 I_0 \omega a^2 \sin(\omega t)}{2 \sqrt{(x-b)^2 + y^2}} \tag{1.1.1.36}$$

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```
Fourier series and transform
```

```
Fourier series
```

```
Fourier series
```

```
Fourier series
```

```
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      "description": "Optional. Specifies the JSON schema to validate against.",
      "enum": [
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```

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$$

```
>>> # Derived Equations Checkpoint
>>> # Verify derived equations from Lagrangian
>>> # Lagrangian: L = (1/2)*m*(dx/dt)**2 - (1/2)*k*x**2
>>> m, k = symbols('m k')
>>> x = Function('x')(t)
```

Step 1: Define the Lagrangian

```
>>> L = (1/2)*m*diff(x, t)**2 - (1/2)*k*x**2
>>> print("Lagrangian:")
Lagrangian:
>>> pprint(L)
```

$$m \left(\frac{d}{d t} x(t)\right)^{2} - k x^{2}$$

```
>>>
```

Step 2: Compute momenta

```
>>> p = diff(L, diff(x, t))
>>> print("Momentum:")
Momentum:
>>> pprint(p)
```

$$m \frac{d}{d t} x(t)$$

```
>>>
```

Step 3: Compute force

```
>>> F = diff(L, x)
>>> print("Force:")
Force:
>>> pprint(F)
```

$$-k x(t)$$

```
>>>
```

Step 4: Verify relation

```
>>> print("Verify: p = m*dx/dt")
Verify: p = m*dx/dt
>>> print("Expected: F = -k*x")
Expected: F = -k*x
>>>
```

Step 5: Derive equation of motion

```
>>> eq = Eq(diff(p, t), F)
>>> print("Equation of motion:")
Equation of motion:
>>> pprint(eq)
```

$$m \frac{d^{2}}{d t^{2}} x(t) = -k x(t)$$

```
>>>
```

Step 6: Solve for acceleration

```
>>> a = solve(eq, diff(x, t, 2))[0]
>>> print("Acceleration:")
Acceleration:
>>> pprint(a)
```

$$\frac{k x(t)}{m}$$

```
>>>
```

Step 7: Verify dimensions

```
>>> # [M][L]/[T]^2 = [M]/[T]^2 * [L]
>>> # [M][L]/[T]^2 = [M][L]/[T]^2
>>> print("Dimensional analysis:")
Dimensional analysis:
>>> print("[M][L]/[T]^2 = [M]/[T]^2 * [L]")
[M][L]/[T]^2 = [M]/[T]^2 * [L]
>>> print("[M][L]/[T]^2 = [M][L]/[T]^2")
[M][L]/[T]^2 = [M][L]/[T]^2
>>>
```

Step 8: Check consistency

```
>>> print("Consistency check:")
Consistency check:
>>> print("Lagrangian → momentum → force → equation → acceleration")
Lagrangian → momentum → force → equation → acceleration
>>>
```

Step 9: Summary

```
>>> print("Derived equations from Lagrangian:")
Derived equations from Lagrangian:
>>> print("1. Lagrangian: L = (1/2)*m*(dx/dt)**2 - (1/2)*k*x**2")
1. Lagrangian: L = (1/2)*m*(dx/dt)**2 - (1/2)*k*x**2
>>> print("2. Momentum: p = m*dx/dt")
2. Momentum: p = m*dx/dt
>>> print("3. Force: F = -k*x")
3. Force: F = -k*x
>>> print("4. Equation: m*d^2x/dt^2 = -k*x")
4. Equation: m*d^2x/dt^2 = -k*x
>>> print("5. Acceleration: a = k*x/m")
5. Acceleration: a = k*x/m
>>>
```

Step 10: Exit

```
>>> exit()
```

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be a compact subset of $\mathbb{R}^d$ for $d \in \mathbb{N}$ and $\mathcal{Y} = \{0, 1\}$ be a finite **Boolean Hypothesis** space.

Let $\mathcal{H} \subseteq \{h : \mathcal{X} \to \mathcal{Y}\}$ be a finite set of hypothesis functions.

Let $\mathcal{D}$ be a distribution over $\mathcal{X} \times \mathcal{Y}$.

2.1 Empirical Risk Minimization

Let $\mathcal{D}_n$ denote a sample of size n.

Lemma 2.1.1 (Hoeffding's Inequality)

Let $X_1, \ldots, X_n$ be independent random variables with $a \leq X_i \leq b$ for all i. Then for $\varepsilon > 0$:

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_i] \geq \varepsilon\right] \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.1.1}$$

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_i] \leq -\varepsilon\right] \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.1.2}$$

Proof.

Let $S_n = \sum_{i=1}^{n} X_i$. Then $\mathbb{E}[S_n] = \sum_{i=1}^{n} \mathbb{E}[X_i]$. Let $\mu = \mathbb{E}[S_n]$. Let $\varepsilon > 0$. Then:

$$\Pr\left[S_n - \mu \geq n\varepsilon\right] = \Pr\left[\exp(\lambda(S_n - \mu)) \geq \exp(\lambda n \varepsilon)\right] \tag{2.1.1.3}$$

$$\leq \frac{\mathbb{E}[\exp(\lambda(S_n - \mu))]}{\exp(\lambda n \varepsilon)} \tag{2.1.1.4}$$

$$= \frac{\prod_{i=1}^{n} \mathbb{E}[\exp(\lambda(X_i - \mathbb{E}[X_i]))]}{\exp(\lambda n \varepsilon)} \tag{2.1.1.5}$$

$$\leq \frac{\prod_{i=1}^{n} \exp\left(\frac{\lambda^2 (b-a)^2}{8}\right)}{\exp(\lambda n \varepsilon)} \tag{2.1.1.6}$$

$$= \exp\left(\frac{\lambda^2 n (b-a)^2}{8} - \lambda n \varepsilon\right) \tag{2.1.1.7}$$

$$\leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.1.8}$$

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_i] \geq \varepsilon\right] \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.1.9}$$

$$\Pr\left[\frac{1}{n}\sum_{i=1}^{n} X_i - \mathbb{E}[X_i] \leq -\varepsilon\right] \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.1.10}$$

Lemma 2.1.2 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.2.1}$$

Proof.

Let $\mathcal{D}_n$ denote a sample of size n. Let $\mathcal{D}_n'$ denote a sample of size n independent of $\mathcal{D}_n$. Let $\varepsilon > 0$. Let $h^* \in \mathcal{H}$ be a fixed hypothesis. Let $\hat{R}(h^*)$ denote the empirical risk of h^* on $\mathcal{D}_n$. Let $\hat{R}'(h^*)$ denote the empirical risk of h^* on $\mathcal{D}_n'$.

Lemma 2.1.3 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.3.1}$$

Proof.

Let $\mathcal{D}_n$ denote a sample of size n. Let $\varepsilon > 0$. Let $h \in \mathcal{H}$. Let $\hat{R}(h)$ denote the empirical risk of h on $\mathcal{D}_n$. Let $R(h)$ denote the true risk of h.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq \sum_{h \in \mathcal{H}} \Pr\left[|R(h) - \hat{R}(h)| \geq \varepsilon\right] \tag{2.1.3.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.3.3}$$

Lemma 2.1.4 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.4.1}$$

Proof.

Let $\mathcal{D}_n$ denote a sample of size n. Let $\delta > 0$. Let $\varepsilon > 0$. Let $h \in \mathcal{H}$. Let $\hat{R}(h)$ denote the empirical risk of h on $\mathcal{D}_n$. Let $R(h)$ denote the true risk of h.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.4.2}$$

$$\leq \delta \tag{2.1.4.3}$$

$$\Rightarrow n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.4.4}$$

Lemma 2.1.5 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.5.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.5.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.5.3}$$

Lemma 2.1.6 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.6.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.6.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.6.3}$$

Lemma 2.1.7 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.7.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.7.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.7.3}$$

Lemma 2.1.8 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.8.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.8.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.8.3}$$

Lemma 2.1.9 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.9.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.9.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.9.3}$$

Lemma 2.1.10 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.10.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.10.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.10.3}$$

Lemma 2.1.11 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.11.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.11.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.11.3}$$

Lemma 2.1.12 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.12.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.12.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.12.3}$$

Lemma 2.1.13 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.13.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.13.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.13.3}$$

Lemma 2.1.14 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.14.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.14.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.14.3}$$

Lemma 2.1.15 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.15.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.15.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.15.3}$$

Lemma 2.1.16 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.16.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.16.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.16.3}$$

Lemma 2.1.17 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.17.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.17.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.17.3}$$

Lemma 2.1.18 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.18.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.18.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.18.3}$$

Lemma 2.1.19 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.19.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.19.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.19.3}$$

Lemma 2.1.20 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.20.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.20.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.20.3}$$

Lemma 2.1.21 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.21.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.21.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.21.3}$$

Lemma 2.1.22 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.22.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.22.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.22.3}$$

Lemma 2.1.23 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.23.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.23.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.23.3}$$

Lemma 2.1.24 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.24.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.24.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.24.3}$$

Lemma 2.1.25 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.25.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.25.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.25.3}$$

Lemma 2.1.26 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.26.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.26.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.26.3}$$

Lemma 2.1.27 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.27.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.27.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.27.3}$$

Lemma 2.1.28 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.28.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.28.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.28.3}$$

Lemma 2.1.29 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.29.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.29.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.29.3}$$

Lemma 2.1.30 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.30.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.30.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.30.3}$$

Lemma 2.1.31 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.31.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.31.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.31.3}$$

Lemma 2.1.32 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.32.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.32.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.32.3}$$

Lemma 2.1.33 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.33.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.33.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.33.3}$$

Lemma 2.1.34 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.34.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.34.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.34.3}$$

Lemma 2.1.35 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.35.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.35.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.35.3}$$

Lemma 2.1.36 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.36.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.36.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.36.3}$$

Lemma 2.1.37 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.37.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.37.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.37.3}$$

Lemma 2.1.38 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.38.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.38.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.38.3}$$

Lemma 2.1.39 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.39.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.39.2}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.39.3}$$

Lemma 2.1.40 (Rademacher Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.40.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.40.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.40.3}$$

Lemma 2.1.41 (Growth Function)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.41.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.41.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.41.3}$$

Lemma 2.1.42 (Symmetrization)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.42.1}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.42.2}$$

$$\leq 2\Pr\left[\sup_{h \in \mathcal{H}} |\hat{R}(h) - \hat{R}'(h)| \geq \frac{\varepsilon}{2}\right] \tag{2.1.42.3}$$

Lemma 2.1.43 (Uniform Convergence)

Let $\mathcal{D}_n$ denote a sample of size n.

$$\Pr\left[\sup_{h \in \mathcal{H}} |R(h) - \hat{R}(h)| \geq \varepsilon\right] \leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.43.1}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.43.2}$$

$$\leq 2|\mathcal{H}| \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right) \tag{2.1.43.3}$$

Lemma 2.1.44 (Sample Complexity)

Let $\mathcal{D}_n$ denote a sample of size n.

$$n \geq \frac{(b-a)^2}{2\varepsilon^2} \ln\left(\frac{2|\mathcal{H}|}{\delta}\right) \tag{2.1.44.1}$$

$$\leq \frac{(b-a)^2}{2\varepsilon^

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